



Intelligent layout generation based on deep generative models: A comprehensive survey

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ABSTRACT

Layout generation, as a crucial stage in Artificial Intelligence-Generated Content (AIGC) task information presentation generation, is a fascinating and significant research topic. Automatically generating layouts intrigued the computer vision community dating back to the 1990s, and the advancement was sluggish for a time due to the inherent complexity of the issue. Thanks to the success of deep learning, layout generation is leading to a surge in both academic literature and business applications recently. Yet a systematic and comprehensive literature review of this domain is still missing. In this study, we provide a comprehensive overview of the state-of-the-art layout generation methods. Primarily, we discuss the problem of layout generation and associated background knowledge including layout representation, layout aesthetics, and generative models. Subsequently, recent advances in the field of intelligent layout generation are organized and summarized from three perspectives: innovations, models & algorithms, and applications. Furthermore, the paper outlines the datasets and evaluation methodology, both qualitative and quantitative, and briefly analyzes the selection of them for specific tasks. To sum up, we conclude the current challenges of intelligent layout generation and suggest possible directions for future endeavors.

1. Introduction

AIGC refers to content that AI automatically creates to meet the personalized requirements of users. The aim of AIGC is to streamline the content creation process and make faster production of high-quality content more accessible to average users. This emerging trend in Artificial Intelligence (AI) technology signifies a shift from the traditional role of AI in perceiving and understanding the world towards generating and creating it. The content generation began with Professional-Generated Content (PGC) and later evolved into User-Generated Content (UGC). With the advancements in Deep Learning technology, AI-assisted User-Generated Content (AIUGC) and AIGC have emerged successively. As content generation mode continues to upgrade, the volume of generated content is increasing rapidly, as illustrated in Fig. 1. Based on the integration of AI technologies such as generative algorithms, pre-trained models, and multimodality, AIGC is versatile in generating various forms of content. One example of this is the large language model (LLM) ChatGPT [1] which has revolutionized the field of human-computer dialogue. ChatGPT is able to engage in high-level EQ dialogues and generate code, scripts, and even novels.

Additionally, Stable Diffusion [2] and DALL-E-2 [3] are state-of-the-art Text-to-Image models that are capable of creating high-resolution images based on prompts. The focus of this survey is on intelligent layout generation, which is an increasingly important research topic in the graphics community as the ability to generate layouts manually can hardly meet the demand of the era of big data, especially the huge demand for advertising graphic design with the rapid development of e-commerce. Thus, it is a research area with immense potential in AIUGC and AIGC.

Layout generation [4] refers to the process of determining the positions, sizes, and some other transformations (flipping, cropping, rotating, etc.) of the compositional primitives that are displayed in a meaningful information arrangement. The term presentation is used to indicate the visual materials with layout attributes to be manipulated. A high-quality layout contributes a lot to information presentation, which enables it to guide people's attention and enhances visual attractiveness. However, automatically generating diverse layouts that can fully convey information and enhance visual attractiveness remains

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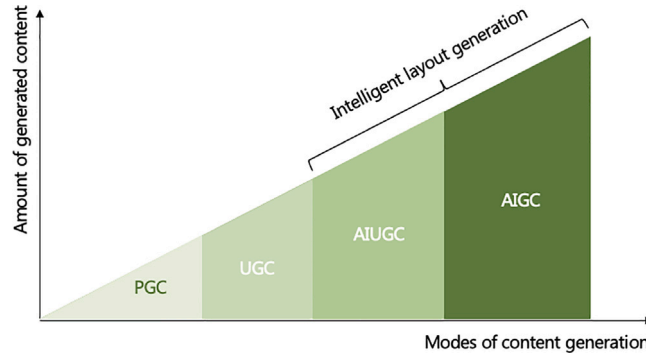


Fig. 1. The development of the mode of content generation.

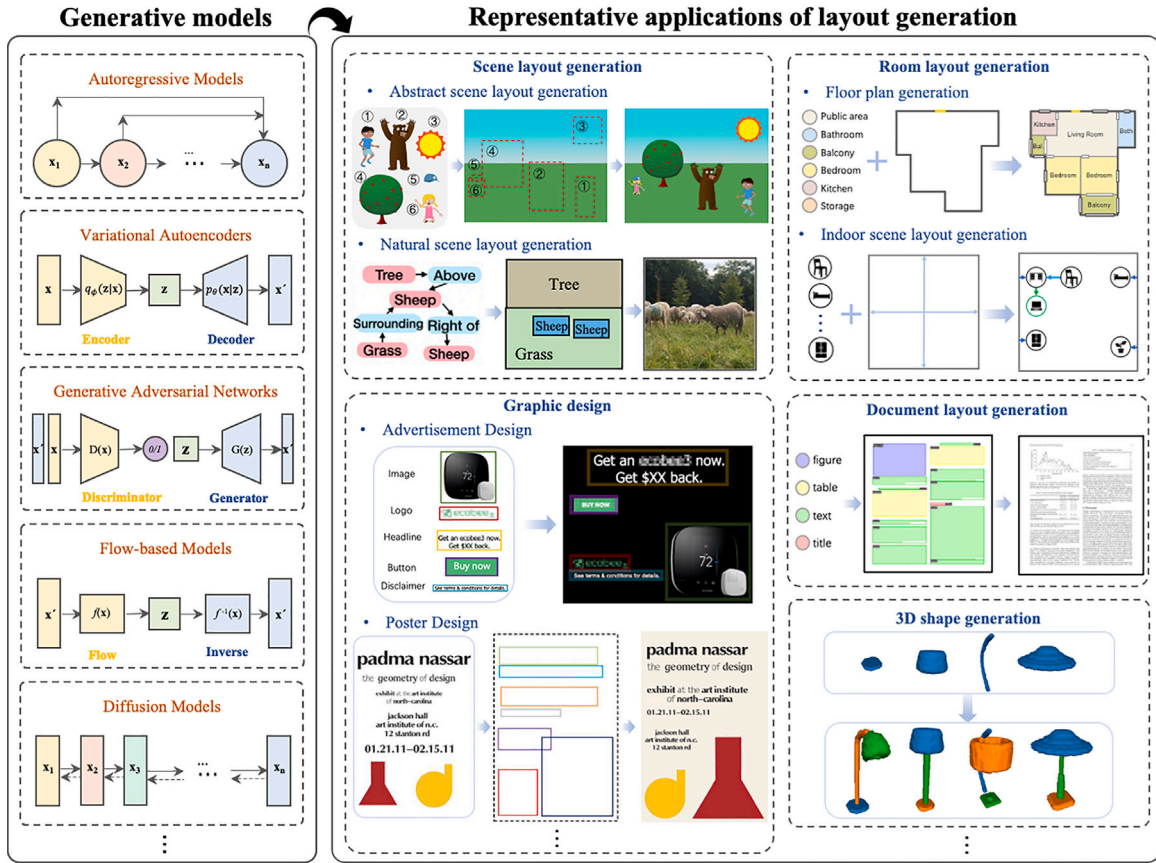


Fig. 2. The generative models and representative applications in layout generation.

a complex challenge, since both the rules of generating spatially and semantically reasonable layouts and aesthetic measurements are difficult to model. Thus, most layout design works are still conducted by humans, which is expensive and time-consuming.

With unprecedented development in deep learning and AI gradually permeating digital content creation, intelligent layout generation has become a research area of interest. Intelligent layout generation means using algorithms and models to automatically either generate layouts or complete partial layouts, which aims to promote the efficiency and quality of design work combined with the rules of artistic creativity. This is an interdisciplinary study of computer vision, computational aesthetics [5], and artificial intelligence. Various deep learning-based generative models can provide theoretical and model support for exploring the applications in this field, such as autoregressive models like Transformer [6], variational auto-encoder (VAE) [7,8], generative adversarial networks (GAN) [9], flow-based model [10], diffusion

model [11], etc. (as shown in Fig. 2 (left)). To get visual-appealing and diverse layouts that can efficiently communicate information, it is necessary to take the contents to be presented (i.e., the compositional primitives), into account. The compositional primitives are different depending on the formats of information presentation which can be scene layout generation, room layout generation, graphic design, document layout generation, 3D shape generation, etc., as shown in Fig. 2 (right). The principles that need to be observed differ from different types of compositional primitives accordingly. For example, in the scene layout generation task, a set of scene elements need to be laid out, and the properties of scene elements and the relationship between them determine the plausibility of the layout; in the graphic design task, it is the textual and image primitives need to follow certain design principles and aesthetic rules; while the compositional primitives in the document layout generation task are titles, figures, tables, captions, text paragraphs, etc., and strict format and readability of the document are supposed to be focused.

Many researchers have been engaging in exploring layout generation problems from different perspectives. However, few review articles have summarized these efforts. Literature [4] reviews and analyzes the automated layout techniques before 2001, mainly the traditional techniques used by research systems. To the best of our knowledge, a systematic comprehensive survey dealing with the recent progress in layout generation is still missing. Thus, we aim to provide an overview of the intelligent layout generation methods in presenting the information. To focus on this domain, this paper will not involve the issues in the field of graph layout [12], Very-large-scale integration (VLSI) [13], and clothing manufacturing (costume design) [14]. Although these fields of study also include the research direction “layout generation”, these layouts are designed to meet their specific requirements (e.g., to minimize chip area in VLSI), rather than to make generated results understandable or visually appealing to humans. Our contributions are threefold. First, we investigate, classify and summarize recent advances in the field of intelligent layout generation. Second, we introduce the datasets and evaluation methods for layout generation issues and summarize the application contexts, which will bring huge convenience to researchers interested in this area. Third, we summarize current challenges in this field and propose possible directions in future work.

The rest of the survey is organized as follows. Section 2 introduces the background of the research on automated layout generation. A taxonomy of the layout generation models from variant aspects is proposed in Section 3. Section 4 reviews the layout generation methods following three perspectives: layout representation, models & algorithms, and evaluation methodology. Section 6 concludes this paper and discusses several open issues that researchers might find intriguing.

2. Background

2.1. Layout representation

The first and most important issue of automatically generating layouts is modeling layouts. A layout is not simply a set of compositional primitives but also composed of the relationships between the components. Layouts usually employ both spatial and abstract relationships. Spatial relationships are closely related to the geometric presentation of the components. By imposing spatial relationships, layouts are more reasonable and visually compelling. Taking document layout as an example, a block of text is supposed to occupy a certain region of the canvas so that all blocks can be aligned making it a neat work. Abstract relationships are high-level relationships that are often associated with semantic information. For instance, the caption of a figure should be relatively close to the figure; a title is important and expected to attract more attention. Since abstract relationships do not specify the position and size of primitives of the layout, generally the first step to express abstract relationships is to turn them into spatial relationships, and how to realize this transformation is another important practical issue.

Intuitively, the relationships in a layout can be represented as a set of constraints by defining handcraft rules. Like traditional layout generation methods, most practices enforce the alignment and fix the number of primitives. However, these rules are too subjective to define explicitly and can hardly extend to different layout tasks. Also, the layout properties can hardly be provided initially as an image or a document structure since they are a result of the arrangement of components and the components with the corresponding relationships are essential for modeling. Recently, most methods for intelligent layout generation use Bounding-Boxes (bbx) to represent the layout elements (as shown in Fig. 2, the wireframes in scene layout generation and graphic design, and the colored blocks in document layout denote the bbx). To deal with layout generation tasks practically, researchers define the layout input and output of the models in different formats so that the layout for specific tasks can be modeled and processed. The input formats can be labels, geometrical parameters of layouts, layout relationship graphs, and multimodal information, etc. The multimodal information

in layout generation issue involves several of various modals including images, labels, keywords, attributes, relationship graph, etc. (as summarized in Table 1). The output is mainly represented as the bbx information (i.e., coordinates, scaling factors, transformation parameters, etc.), while there are also some other formats of the output such as images and layout denoted by semantic segmentation annotations. The representation formats of the input and output in different layout generation applications are summarized in Table 1.

2.2. Layout aesthetics

Aesthetic is a psychology that focuses on two opposing traditions [41], the objectivist and the subjectivist. The former emphasizes object properties as the source of aesthetics, including symmetry, proportion, and order. While the latter maintains that aesthetics is based on individual and subjective responses. Therefore, aesthetics is also defined as a discipline of perceptive and cognitive research. Birkhoff [42] proposes a mathematical formulation of the aesthetic measure according to the aesthetic demand for ‘unity in variety’ as follows:

$$M = \frac{O}{C}, \quad (1)$$

where M denotes the aesthetic measure, C is the complexity of the object, and O is the order of the object which is important since the object is characterized by a certain harmony, symmetry, or order. Although a standard mathematical formulation of the aesthetic measure is not defined and tested, the formula is regarded as the prototype of computational aesthetics.

Computational aesthetics [5] is proposed to achieve computable aesthetic decision-making and aesthetic generation in the context of computer science. The main branch of computational aesthetics is computational image aesthetics, which aims to enable computers to independently perform quantitative analysis, calculation, and evaluation of the aesthetic measure of images. Computational image aesthetics involve color, hue, layout, etc. Savakis [44], Datta [45] et al. point that, among them, layout plays the most important role in determining the attractiveness and the quality of an image. Aesthetic principles are sophisticated, vague in many descriptions, and difficult to build computational models directly. The layout aesthetic measure rules are studied as the theories for designing optimization losses to promote the layout generation models and evaluation metrics to assess the performance of the models. According to practical demand and effect, various rules are adopted for different context applications. Based on the prior work related to aesthetic assessment [46–48], quantifying principles like alignment, overlap, and visual balance significantly impact the readability and visual attractiveness of the generated layout presentation and the effect of removing these rules from layout generation is illustrated in Fig. 3. More losses and evaluation metrics adopted in layout generation methods are summarized and introduced in detail in Section 5.

2.3. Generative models

Research on generative models has been a hot topic for decades, since to truly understand computer vision systems, not only recognizing images but generating images is practically necessary. The basic idea of generative models is to take a collection of training examples consisting of samples drawn from a distribution p_{data} and learn to represent a probability distribution p_{model} that explains where the trending examples come from.

Deep generative model families are designed to capture the likelihood distribution of concerning variables and can be divided into three categories: explicit density models (parametric methods), implicit density models (non-parametric method), and variant models. Explicit density models realize generation by explicitly formulating the likelihood functions, as shown in Fig. 4, which can be tractable or



Fig. 3. Effect of removing various rules from layout generation [43].

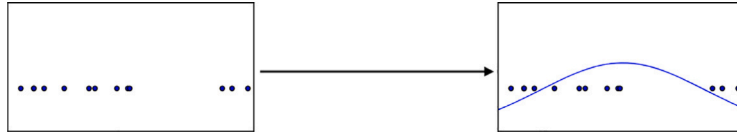


Fig. 4. Explicit density models estimate p_{model} explicitly by taking a training set of examples drawn from a real data distribution p_{data} and returning an estimate of that distribution. This figure illustrates the process for a collection of samples of one-dimensional data and a Gaussian model.

Source: Image from [49]

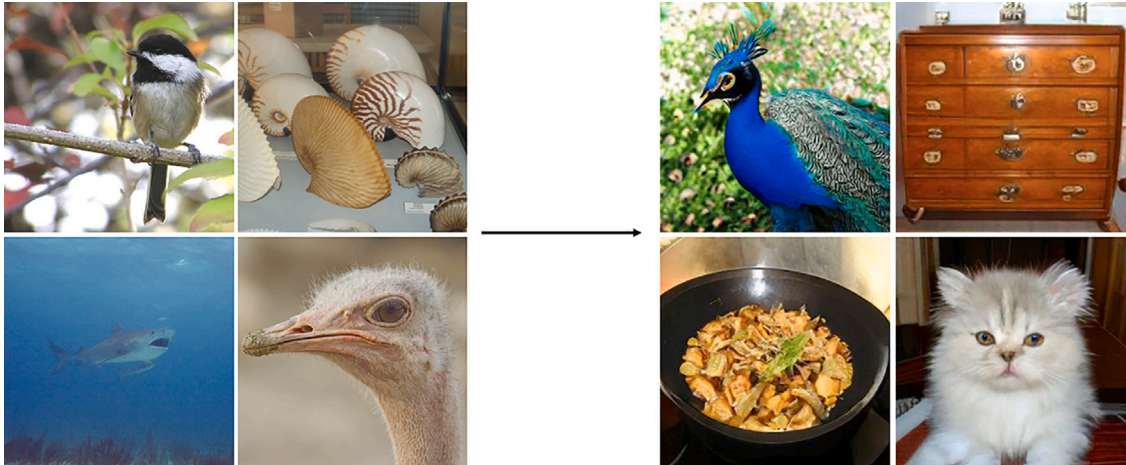


Fig. 5. The illustration of the process of generating samples from the model distribution. The training examples are sampled from the ImageNet [50] dataset as shown on the left and examples generated by an implicit density model, BigGANs [51], as shown on the right.

Table 1
Representation in layout generation.

Input	Methods	Output	Applications					
			(1)	(2)	(3)	(4)	(5)	(6)
Geometrical parameters	LayoutGAN [15]	Bbx information	✓	✓		✓		
	LayoutTransformer [16]	Bbx information	✓	✓		✓	✓	
	VTN [17]	Bbx information		✓	✓	✓		
	Hu et al. [18]	Bbx information		✓				
Multimodal information	ContentGAN	Image		✓				
	Pang et al. [19]	Image		✓				
	L-CGAN [20]	Bbx information	✓	✓		✓		
	Zhou et al. [21]	Bbx information		✓				
	Qiao et al. [22]	Layout ^a						✓
	BLT [23]	Bbx information	✓	✓	✓	✓		
	Play [24]	Layout ^b	✓	✓	✓	✓		
	Li et al. [25]	Image	✓					
	Kikuchi et al. [26]	Bbx information		✓		✓		
	Graph2Plan [27]	Floorplan			✓			
Layout relationship graph	House-GAN++ [28]	Layout ^c			✓			
	NDN [29]	Bbx information		✓				
	Paraet al. [30]	Bbx information			✓			
	House-GAN [31]	Layout ^c			✓			
	Di et al. [32]	Scene instantiation			✓			
Labels	PlanIT [33]	Bbx information			✓			
	LayoutVAE [34]	Bbx information	✓					
Text embedding	Hong et al. [35]	Image	✓					
Scene graph (SG) ^d	LT-Net [36]	Bbx information	✓					
Labeled bounding boxes	READ [37]	Bbx information				✓		
Image	Ritchie et al. [38]	Bbx information			✓			
Depth map	3D-PRNN [39]	Primitive set					✓	
Point cloud	ComplementMe [40]	Point cloud					✓	

(1) Scene layout.

(2) Graphic design.

(3) Room layout.

(4) Document layout.

(5) 3D shapes.

(6) New-view layout synthesis.

^aIn the format of semantic segmentation annotations.

^bIn the format of latent representations.

^cIn the format of the tightest axis-aligned rectangle.

^dSG will be converted into the input format required by the transformer network (sequential data in random order and separate triplets with a special token of “SEP”).

approximate. Theoretically, all generative models can be formulated in an explicit format, but it could be difficult to directly calculate the probability distribution in practice, even in an approximate way. Thus, it is of great significance to use non-parametric methods, namely implicit density models, to sample from the data and capture the distribution p_{model} implicitly, as shown in Fig. 5. Combining the two different ideas of explicit density models and implicit density models, some variant models, like flow-based methods, are proposed. According to the classification, we provide a taxonomy of deep generative model families as illustrated in Fig. 6.

Among these models, the most commonly used backbones for layout generation tasks are VAE, GAN, Transformer, and Diffusion models. The following introduction will focus on these deep generative models.

2.3.1. VAE

The Variational Autoencoder (VAE) [7,8] is an instance of generative models using learned approximate inference and trained with gradient-based methods [52]. The input $\mathbf{x}$ is mapped into a distribution $p_{model}(\mathbf{z})$ from which $\mathbf{z}$ is sampled and fed to a differentiable generative network $g(\mathbf{z})$. During the training process, the encoder $q(\mathbf{z}|\mathbf{x})$ works as an approximate inference network for capturing $\mathbf{z}$, and $p_{model}(\mathbf{x}|\mathbf{z})$ is regarded as a decoder as shown in Fig. 2 (left ‘Variational Autoencoders’). It is trained by maximizing the variational lower bound $\mathcal{L}(q)$ associated with $\mathbf{x}$, which can be formulated as follows:

$$\begin{aligned}\mathcal{L}(q) &= \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z}|\mathbf{x})} \log p_{model}(\mathbf{z}, \mathbf{x}) + H(q(\mathbf{z}|\mathbf{x})) \\ &= \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z}|\mathbf{x})} \log p_{model}(\mathbf{x}|\mathbf{z}) - D_{KL}(q(\mathbf{z}|\mathbf{x}) \parallel p_{model}(\mathbf{z}))\end{aligned}$$

$$\leq \log p_{model}(\mathbf{x}). \quad (2)$$

Here, $\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z}|\mathbf{x})} \log p_{model}(\mathbf{z}, \mathbf{x})$ is recognized as a joint log-likelihood of the visible and hidden variables under the approximate posterior over the latent variables, and $H(q(\mathbf{z}|\mathbf{x}))$ is the entropy of the approximate posterior that encourages the variational posterior to place high probability mass on many $\mathbf{z}$ values that could have generated $\mathbf{x}$, rather than collapsing to a single point estimate of the most likely value, where q is generally chosen to be a Gaussian distribution. $D_{KL}(q(\mathbf{z}|\mathbf{x}) \parallel p_{model}(\mathbf{z}))$ tries to make the approximate posterior distribution $q(\mathbf{z}|\mathbf{x})$ and the model prior $p_{model}(\mathbf{z})$ approach each other. Traditional approaches to variational inference and learning infer q via an optimization algorithm. These approaches are slow and often require the ability to compute $\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z}|\mathbf{x})} \log p_{model}(\mathbf{z}, \mathbf{x})$ in closed form. The main idea behind the variational auto-encoder is to train a parametric encoder that produces the parameters q . As long as $\mathbf{z}$ is a continuous variable, we can then back-propagate through samples of $\mathbf{z}$ drawn from $q(\mathbf{z}|\mathbf{x}) = q(\mathbf{z}; f(\mathbf{x}; \theta))$ to obtain a gradient with respect to θ . Learning then consists solely of maximizing $\mathcal{L}$ with respect to the parameters of the encoder and decoder.

The mechanism of VAE is supported by Bayesian inference so it has an explicit probabilistic interpretation. It can generate smooth and continuous data, which is applicable to interpolation tasks. However, the images generated via VAE tend to be blurry and not on par with GANs, due to the neglect of the unfrequent features in latent variable space and the punishment from KL Divergence. In addition, the practice of mapping the high-dimensional input data into low-dimensional latent

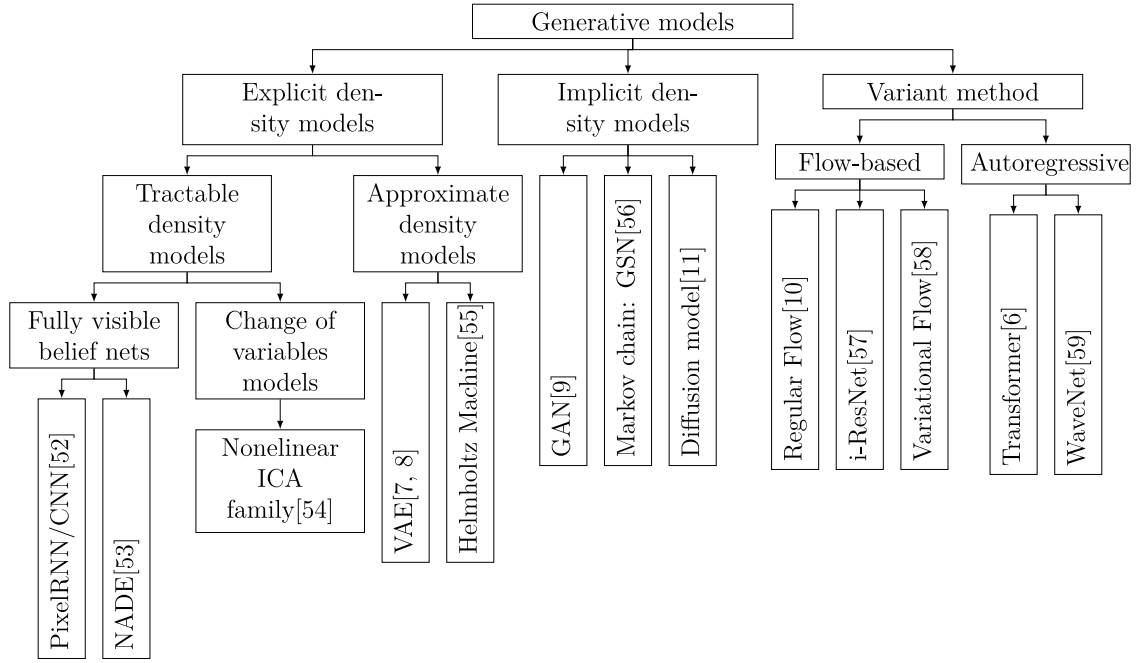


Fig. 6. Classification of Deep Generative Models.

variable space may cause information loss and inadequate representing capacity.

2.3.2. GAN

Generative adversarial networks (GANs) [9] are a generative method based on differentiable generator networks. GANs are estimated via an adversarial process, where two models are trained simultaneously: a generative model G is trained to capture the real data distribution, and a discriminative model D aims to estimate the probability that a sample is drawn from the training data rather than G . The goal of G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. During learning, both players attempt to maximize their own payoff. To learn p_g , the distribution generated over real data x by G , a prior on input noise variables $p_z(z)$ is defined, and mapped to data space as $G(z)$, where G is a differentiable function. The output of D is represented as $D(x)$, which is a single scalar indicating the probability of assigning the correct label to both training examples p_{real} and samples from p_g . Maximizing $D(x)$ and minimizing $\log(1 - D(G(z)))$ are trained simultaneously via the objective function $V(G, D)$:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x \sim p_{real}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log (1 - D(G(z)))]. \quad (3)$$

Finally, the two-player minimax game reaches a convergence:

$$G^* = \arg \min_G \max_D V(G, D). \quad (4)$$

The whole process can be illustrated as shown in Fig. 2 (left 'Generative adversarial networks'). GAN can use a large number of unlabeled samples to learn from latent space to data space and return a high dimensional non-linear map, which is important for unsupervised and semi-supervised learning tasks such as image generation [53], style transfer [54], super-resolution [55], etc. GAN-based methods can generate high-resolution images, and process large-scale and high-dimensional data by generating from high-dimensional stochastic noise rather than mapping the input into low-dimensional latent space. Nevertheless, compared with VAE, GAN does not have an explicit probabilistic interpretation. Moreover, GAN is hard to train because it struggles to balance the competitive relationship between the generator and the discriminator as well as avoid problems such as mode collapse and gradient vanishing.

2.3.3. Transformer

Transformer [6], an autoregressive model (see Fig. 2 Autoregressive Models) was originally proposed for natural language processing (NLP) task machine translation. Later works show that Transformer-based models can achieve state-of-the-art performances on various tasks, not only in the field of NLP but also in computer vision (CV) [56]. Unlike VAE and GANs, Transformer is a sequence-to-sequence model [57] designed for processing sequential data. The vanilla Transformer [6] consists of an encoder and a decoder with a stack of identical blocks. Each transformer block is mainly composed of a multi-head attention module, a feed-forward neural network (FFN), a residual connection [58], and layer normalization [59]. Notably, the decoder additionally adopts cross-attention modules between the multi-head self-attention modules and the position-wise FFNs.

The overall architecture of the vanilla Transformer is demonstrated in Fig. 7. The attention function (see Fig. 7 right) is computed by the attention mechanism with Query-Key-Value (QKV) model, as follows:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V = AV \quad (5)$$

Where d_k denotes the dimension of keys (or queries), and $A = softmax(\frac{QK^T}{\sqrt{d_k}})$ is often called an attention matrix. To counteract the gradient vanishing problem of the softmax function, the dot-products of queries and keys are scaled by $\sqrt{d_k}$.

Instead of simply performing a single attention function, Transformer uses multi-head attention (see Fig. 7 middle) that allows the model to jointly attend to information from different representation subspaces at different positions.

$$MultiHeadAttn(Q, K, V) = Concat(head_1, head_H)W^O \quad (6)$$

Where $head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$, the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{model} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{model} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{model} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{model}}$.

Each layer in the encoder and decoder of Transformer contains a fully connected FFN and operates separately and identically on each position with ReLU activation.

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2, \quad (7)$$

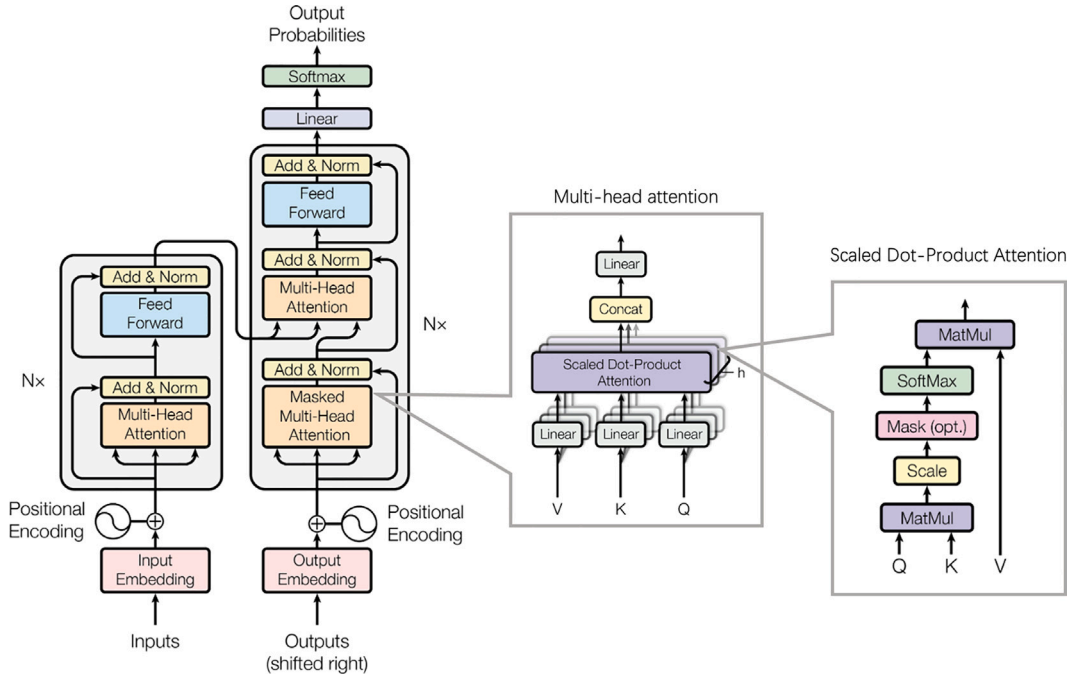


Fig. 7. The model architecture of Transformer.

Source: Image source: Figure 1 & 2 in [6]

To build a deep model, Transformer employs a residual connection [58] around each module. A layer normalization [59] is followed after the residual connection. The output of these operations can be written as:

$$\text{LayerNorm}(X + \text{Attention}(X)). \quad (8)$$

Transformer does not need recurrent or convolutional structures to deal with sequential data, but directly uses the self-attention mechanism to establish global dependencies, thus it can handle long sequences and parallel computation. However, Transformer costs a lot of computational resources and data to train and relies on some tricks to avoid overfitting and gradient explosion. The self-attention mechanism can capture the dependencies between different positions in the sequence, but its specific working function is not easy to understand and analyze.

2.3.4. Diffusion models

Diffusion models, inspired by nonequilibrium thermodynamics, have shown state-of-the-art performance in density estimation [60] and in sample quality [61]. The core idea of diffusion models is to define a Markov chain of diffusion steps that slowly add random noise to data (called the forward process or diffusion process) and then learn to reverse the diffusion process to construct desired data samples from the noise (called the reverse process). The whole pipeline is demonstrated in Fig. 8. Given a data point sampled from a real data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$, a small amount of Gaussian noise is added to the sample in T steps during the forward diffusion process, producing a sequence of noisy samples $\mathbf{x}_1, \dots, \mathbf{x}_T$. The step sizes are according to a variance schedule $\beta_1, \dots, \beta_T$:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t I). \quad (9)$$

Note that the training is performed by optimizing the usual variational bound on negative log-likelihood where β_t can be learned by reparameterization [7] or held constant as hyperparameters.

If the above process can be reversed and sampled from $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$, the true sample will be able to recreate from a Gaussian noise input

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Note that if β_t is small enough, $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ will also be Gaussian. The reverse diffusion process is defined as a Markov chain with learned Gaussian transitions starting at $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$:

$$p_\theta(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t), \quad (10)$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sum_\theta(\mathbf{x}_t, t)).$$

Efficient training works by optimizing random terms of diffusion loss L with stochastic gradient descent (See Appendix A [11] for details).

Diffusion models have the advantage of being trained with a simple maximum likelihood objective and standard optimizers, avoiding adversarial training which can be unstable, difficult to tune, and prone to mode collapse. They are also flexible for different generative tasks, as they can use different types of encoders, decoders, priors, and diffusion processes. However, diffusion models have the drawback of being slower to sample than GANs, as they need to run multiple steps in the denoising process to generate a single sample. Another challenge is the risk of posterior collapse, where the latent variables become independent of the input data and the model becomes a simple autoregressive model, which can reduce the diversity and expressiveness of the generated data.

Application-oriented comparison of VAE, GAN, Transformer, and Diffusion model is summarized based on the previous introduction [6–9,62] as shown in Table 2. Noted that for generative models, there is a common disadvantage, that is, difficult to evaluate empirically. Among the Pros and Cons, sample quality is used to evaluate the performance of image generation tasks, which may not be appropriate for layout generation. By designing and modeling the proper input format for layout, these generative models can also be applied to layout generation issues. The layout generation methods based on different models will also leverage their strengths or face their limitations, and the specific approaches will be introduced in detail in Section 4.

3. A taxonomy of layout generation models

So far, a wide variety of models have been proposed, we explore them from three perspectives: types of models, the content of information presentation, and applications. For comprehensive understanding,

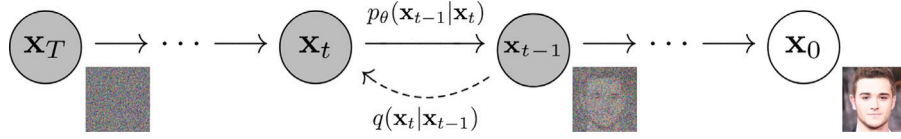


Fig. 8. The whole pipeline of Diffusion models. The Markov chain of the forward/reverse diffusion process of generating a sample by slowly adding/removing noise. Source: Image from [11]

Table 2

Application-oriented comparison of VAE, GAN, Transformer, and Diffusion model.

Model	Pros	Cons
VAE	• High interpretability. • Achieves high data likelihood. • Measurable objective.	• Low sample quality. • Limited approximation to true posterior.
GAN	• Parallel sample generation (fast). • High sample quality. • No Markov chains needed.	• Require to find Nash equilibrium — harder problem than optimizing an objective function. • Difficult to train (instability and possible collapse). • Lack of interpretability.
Transformer	• High sample quality. • Expandability.	• High computational and training data cost. • Lack of interpretability.
Diffusion model	• Straightforward to define. • Efficient to train. • High sample quality. • Expandability.	• Low efficiency (slow) in sampling. • High computational and training data cost. • May suffer from posterior collapse.

Fig. 9 gives an illustration of our taxonomy of layout generation. The model-based explorations include the traditional methods and Deep learning (DL) methods with different backbone networks. The content categories can be roughly divided into content-aware layout generation and content-agnostic layout generation depending on whether the primitives of the layout are provided or not. The main application categories we explore include scene layout, graphic design, document layout, room layout, and other tasks.

Based on our taxonomy, we will discuss more specifically the methods, datasets, and related evaluation methodology in Sections 4 and 5.

4. Layout generation derivations

We organize the following parts by categorizing models into different tasks and analyzing their main ideas, advantages, and disadvantages. For a better demonstration of the derivations and development, we introduce each category according to the timeline and the methods might be naturally divided into methods with and without neural networks.

4.1. Scene layout

The rapid progress and recent successes in image generation [63–67] contribute to frameworks designed for stochastic layout generation. Scene layout generation problems involve abstract scene generation tasks and natural scene generation tasks. The former generates reasonable layouts and a final full scene can be obtained by placing given abstract primitives guided by generated layouts (see Fig. 2 (Abstract scene layout generation)). The latter basically is an image generation issue, and researchers found that using a two-stage approach that generates a scene layout first, then synthesizes a natural scene from a layout utilizing methods proposed by [68,69], etc., can improve the validity and the quality of generated results. Therefore, the study of scene layout generation was introduced to the field of scene generation as the first and very important step.

In this section, we will summarize the recently proposed scene layout generation methods that help improve the performance of abstract scene layout generation and natural scene layout generation.

4.1.1. Abstract scene layout generation

Abstract scene layout generation can be regarded as a simplified layout generation problem since it only needs to synthesize scenes from provided abstract image elements guided by generated layouts (see Fig. 2 (Abstract scene layout generation)) rather than generating a scene pixel-by-pixel as in the natural scene generation task (see Fig. 2 (Natural scene generation)), or filling primitives from different modalities (e.g., text, tables, figures), into generation targets as in document generation and graphic design problem. Indeed, abstract scene generation reveals the core of many generative tasks, that is, generating a reasonable layout.

Li et al. [15] presented LayoutGAN with simplified problem-setting, i.e., generating a layout based on input noise without considering more detailed rules contained in a scene, like count relationships and user requirements. The overall architecture of LayoutGAN is clear and unambiguous as illustrated in Fig. 10. The input of the generator is graphic elements with randomly sampled class probabilities and geometric parameters from Uniform distribution and Gaussian distribution respectively. The stacked relation module refines the embedded features of each element from an encoder considering the semantic and spatial relations between all element pairs. Finally, a decoder decodes the refined features back to class probabilities and geometric parameters. The generated results are fed into the discriminator with a differentiable wireframe rendering layer and a CNN layer for layout optimization. As the opening work in this domain, the generated scenes are relatively simple and contain fewer objects, but this work is significant. In addition to the abstract scene generation, the effectiveness is also validated via various experiments including MNIST digit generation, document layout generation, and tangram graphic design.

Shi et al. [20] proposed a layout generation model based on Conditional Generative Adversarial Networks (L-CGAN), which is a two-stage method that can generate a layout with given element labels through a CGAN-based model and refine the layout following the provided relationship graph via a graph-constraint module. To break the limitation of the fixed input size of the network, a pre-processing algorithm is introduced and implemented before feeding the structured data as input into the model. The competitive performance in diverse data domains, such as handwriting digit layout generation, scene layout generation, and document layout generation, is achieved due to the

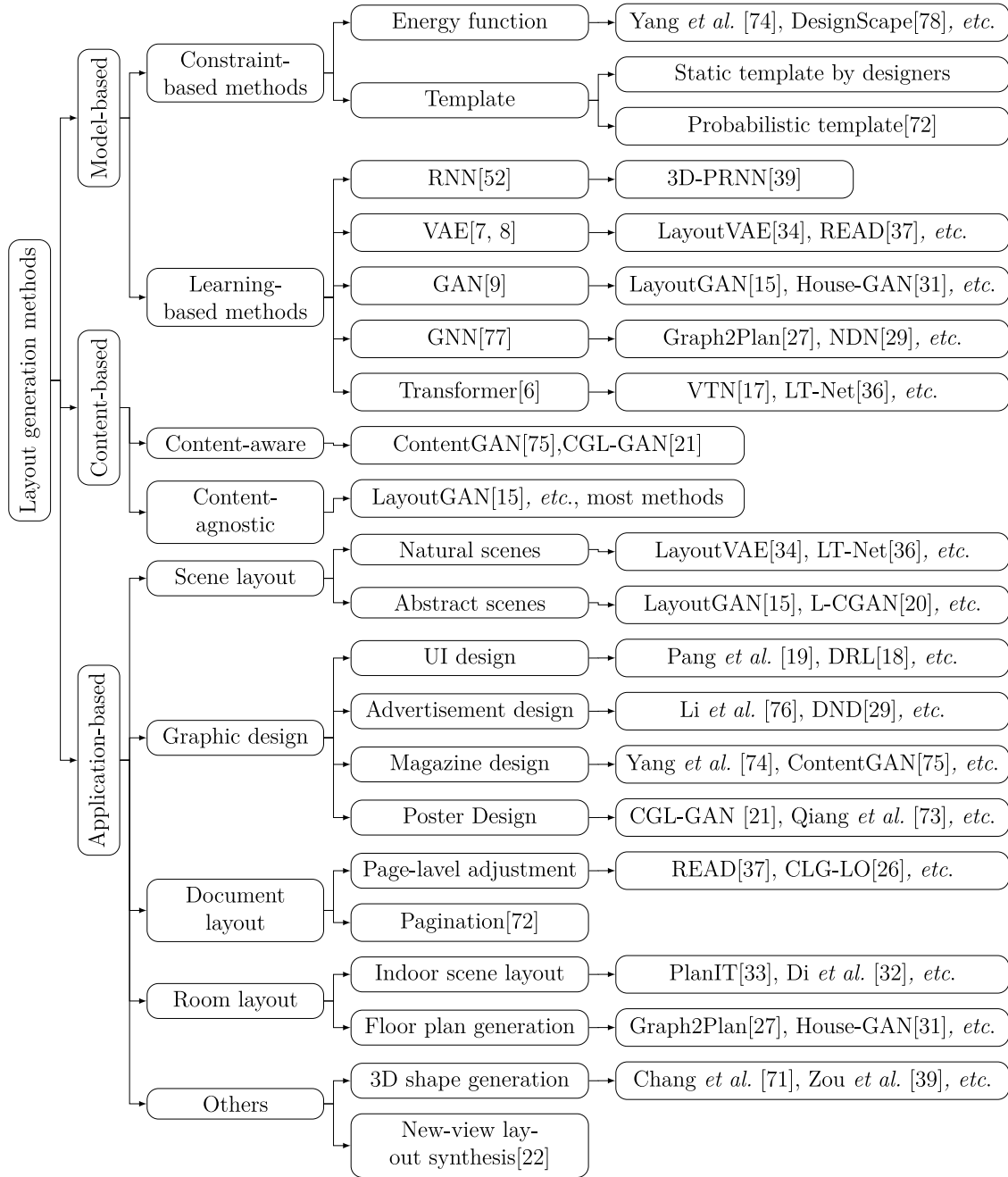


Fig. 9. Taxonomy of Layout Generation Models.

focus on the contextualized relationship, despite that the relationship graph is defined manually instead of extracted automatically.

4.1.2. Natural scene generation

Text-to-image synthesis issue is a hot topic in the computer vision community, and as research progresses the issue can be extended into the natural scene generation problem with more variants including synthesizing images from labels, scene graphs, and sketches. Likewise, existing works for natural scene generation usually synthesize plausible scenes via direct pixel-to-pixel mapping conditioned on provided labels, scene graphs, text descriptions, etc. However, by this means the results are not as good as expected due to the neglect of plausible visual variations within the structure dictated by conditional information.

To enhance the quality of generated results, Hong et al. [35] propose a novel hierarchical approach for text-to-image synthesis which

decomposes the synthesis process into multiple steps in a coarse-to-fine manner (see Fig. 11). Given a text embedding, a coarse layout is first constructed by placing a set of object bounding boxes by the box generator, and further, the object shape inside each bounding box is refined via the shape generator. Conditioned on the inferred semantic outputs from the box and the shape generator, which provides a useful semantic structure of an image matching the text description, a pixel-wise image is finally generated by the image generator. The box generator, shape generator, and image generator employ an autoregressive decoder with LSTM [70], a convolutional recurrent neural network [71], and a convolutional encoder-decoder network [72], respectively. By conditioning image generation on explicit layout prediction, generated images preserve semantic details and highly match the text description.

LayoutVAE [34] proposed by Jyothi et al. also adopts the hierarchical strategy but focuses on layout generation. Jyothi et al. argue that

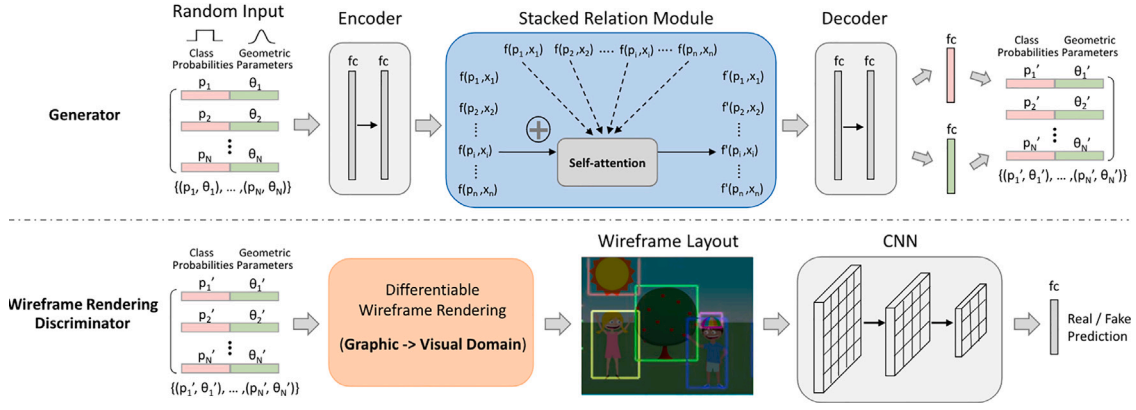


Fig. 10. Overall architecture of LayoutGAN.

Source: Image from [15]

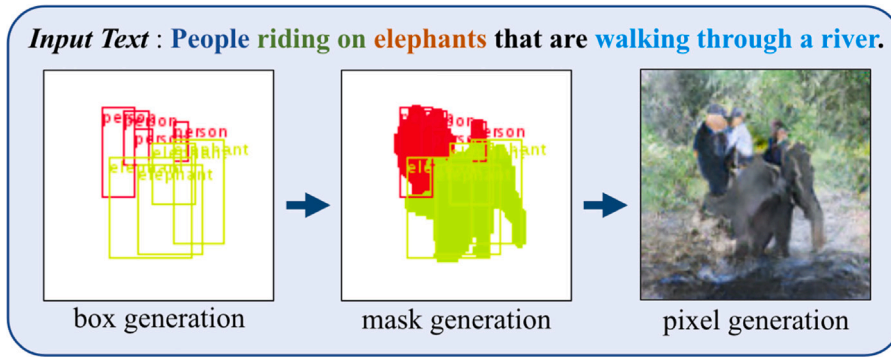


Fig. 11. Overall framework of the model proposed in [35].

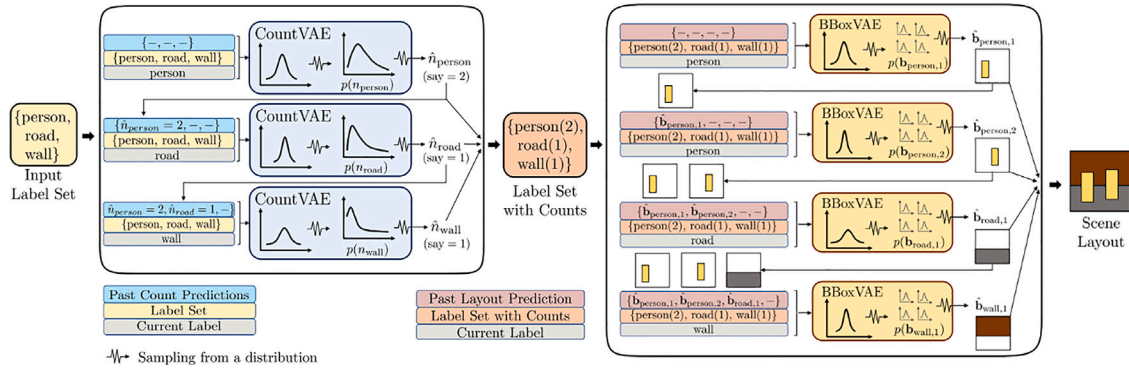


Fig. 12. Overall architecture of LayoutVAE.

Source: Image from [34]

solving this challenging task requires answering two questions: what is the number of objects in each category? and what is the location and size of each object? Naturally, LayoutVAE is decomposed into two models, CountVAE and BBoxVAE, to model the distribution of count and spatial relationships between objects in a scene, respectively. As shown in Fig. 12, it is a two-step approach: first, the number of objects (count) for each label is predicted by CountVAE, then BBoxVAE predicts the bounding box for each object to generate diverse layouts. LayoutVAE is flexible enough to not only realize layout generation but also layout completion, *i.e.*, it can be used to generate a layout or add a single bounding box in an existing image given a new label. To visualize diverse realistic layouts generated by LayoutVAE, Jyothi et al. generate images from the layouts via the model proposed by Zhao et al. [68]. Qualitative visualizations from the experiments show that LayoutVAE can learn intrinsic relationships between objects in scenes.

Noted that the contextualized relations between the objects contained in text sentences may not be properly described when translating text inputs into layouts or images, which would make the generated results less plausible. Therefore, explicit relationship descriptions in a scene are typically required. LayoutVAE takes a set of object labels as inputs, but still might not sufficiently exploit the text descriptions, since labels cannot explicitly describe relationships between objects in scenes. Inspired by [64], Yang et al. [36] use scene graphs (SG) as an intermediate text representation, and propose a LayoutTransformer Network (LT-Net) attempting to infer implicit objects or relationships properly during layout generation. Given a scene graph with objects and relations explicitly presented as input, the encoder in LT-Net encodes the semantic features for exploiting their co-occurrences and implicit relationships to generate conceptually diverse yet plausible layouts. The decoder of LT-Net translates the encoded contextual features

into bounding boxes. With the enforcement of GMM-like distributions and the self-supervised relation consistency, the generated layouts can be spatially diverse. For visualization purposes, the pre-trained LostGAN [73] is employed to convert the generated layout into images. The experiments qualitatively and quantitatively demonstrated the competitive performance of LT-Net over state-of-the-art scene generation methods, including Sg2Im [64], CanonicalSg2Im [74] (SG-to-image methods) and NDN [29] (layout generation method introduced in Section 4.2).

4.1.3. Discussions

Abstract scene layout generation is the initial task of layout generation models based on deep learning, and it can contribute to building datasets for scene understanding and other computer vision tasks. Natural scene layout generation is a novel approach for natural scene synthesis, which provides an intermediate stage to generate scene layout and then synthesize the scene guided by the generated layout. The two-stage scene synthesis strategy improves the plausibility of the generated scene. As the study on natural scene layout generation moves along, research on layout-to-image generation [68,73,75,76] is also promoted. The scene layout generation models aim to generate plausible scene layouts both for abstract and natural scenes and the evaluation metrics of the models are summarized in Section 5.

4.2. Graphic design

The layout is considered the core of graphic design, including UI design [16,19,29,77], advertisement design [29,78], magazine design [79,80], poster design [21,81], etc. A layout can be produced by selecting a template that best fits the given content. However, such a predefined, restrained set of templates can rarely describe the rich variants of graphic design layouts. Although there are studies aiming to present automatic methods for generating templates [81–83], the diagram of the templates generally requires professional prior knowledge and manual effort and provides less flexibility. In recent years, a rising interest in graphic design using layout generation methods has been witnessed in the computer vision community, especially with deep learning technology.

In this section, we will provide a review of how layout generation methods work in a variety of specialized graphic design tasks.

4.2.1. UI design

UI design refers to User Interface design, which is the process of designing the interface and interaction elements of digital products such as software applications, websites, and mobile devices. The demand for UI design has grown significantly in recent years as more and more businesses are investing in digital products. The goal of UI design is to make the product easy to use, visually appealing, and accessible to a wide range of users. In recent years, researchers have aimed to automatically generate functional and visually appealing UI designs and made a lot of attempts and efforts.

Pang et al. [19] propose user attention models to optimize web designs and guide user attention along designer-specified input paths. Todi et al. [77] study the UI design problem based on the observation of the vast untapped potential of optimization methods in design tools and propose a novel approach for incorporating real-time design optimization into a sketching tool to improve hand-drawn layouts according to certain visual design rules. Lee et al. [29] propose a GNN-based deep neural network named the Neural Design Network (NDN), which is capable of generating high-quality design layouts given a set of desired components along with user-specified constraints. The NDN consists of three parts: relation prediction, bounding box prediction, and refinement. The relation prediction module captures the dependency among components in a layout by inputting a graph with partial relations and a latent vector and outputs a graph representing the complete relations. The layout generation module utilizes the predicted graph

as input and generates a preliminary design layout by predicting the bounding boxes for all nodes in the graph iteratively. The refinement module further fine-tunes the layout by fixing the predicted bounding boxes in an iterative manner. The NDN has demonstrated good performance in many real layout-based design cases, including Mobile app UI design, Magazine layout generation, and Advertisement design. The LayoutTransformer presented by Gupta et al. [16] is a transformer-variant framework that leverages the self-attention module to capture contextual relationships between layout primitives and generate layouts in various domains including UI design, 3D shapes, natural scene generation, document layout generation.

Although state-of-the-art Transformer-based methods like LayoutTransformer [16] and VTN [36] can be adjusted to UI design tasks, Kong et al. [23] found they are only effective for unconditional layout generation, where layout elements are generated from random seeds without respecting to specific user requirements (see Fig. 13). Additionally, they have difficulties in conditional generation tasks because of the immutable dependency chain and high latency in decoding. To enable controllable layout generation, Bidirectional Layout Transformer (BLT) is introduced. Unlike existing works [16,36], BLT is designed with bidirectional attention and a parallel decoding algorithm, allowing it to fulfill conditional generation and generate attributes simultaneously in parallel. This enables flexible conditional generation, as well as more efficient inference.

With the explosion of diffusion models [11] as the promising generative framework and their state-of-the-art on image and video generation tasks, researchers are now attempting to introduce diffusion models to the field of layout generation. Cheng et al. [24] build Parametrically Conditioned Layout Generation using Guidelines (PLay), a two-stage model based on latent diffusion models (LDMs) [2]. PLay is able to generate and edit parametrically conditioned layouts in vector graphic space from user-specified requirements for different tasks, including guideline editing, layout inpainting, and completing layout from a partial layout.

4.2.2. Advertisement design

Advertisement design is another important practice of graphic design that aims to promote a product, service, or brand and grab the attention of potential customers. With the emergence of e-commerce, there has been a surge in the demand for advertisement design. The study on automagical advertisement design is of great value in practice.

Li et al. [78] propose an attribute-conditioned Layout GAN dedicated to solving advertisement layout generation, which is an enhanced version of LayoutGAN [15]. The model can generate an attribute conditional graphic layout by forcing attribute conditions on both the generator and the discriminator. Innovatively, three attributes supposed to take into consideration in a professional design are introduced, which are expected area, aspect ratio, and reading order. Concretely, the expected area attributes of bounding boxes are fed to the generator to control the area of the generated elements and trained with Margin Area Loss. The aspect ratio of the generated bounding box of an image element is supposed to be fixed, that is, H/W (H and W are the height and the width of the bounding box of the input image element respectively). The reading order of design elements follows simple heuristics for better information presentation (i.e., the left-to-right and top-to-down), and an Order Loss is introduced to penalize element pairs that have against relative distances to the layout origin with their predefined reading orders during layout adjustment. Hu et al. [18] study on banner design via layout generation mode and present a traditional method transformed adjusting positions of design elements into algorithms separately and a deep reinforcement learning (DRL) [84] based method with two different reward functions for layout generation. They also quantitatively compare the two methods through evaluation metrics designed based on the work of [43], including text alignment, overlap, and visual balance (summarized in Table 4). The experiment turns out that both models can generate banner layouts with a given

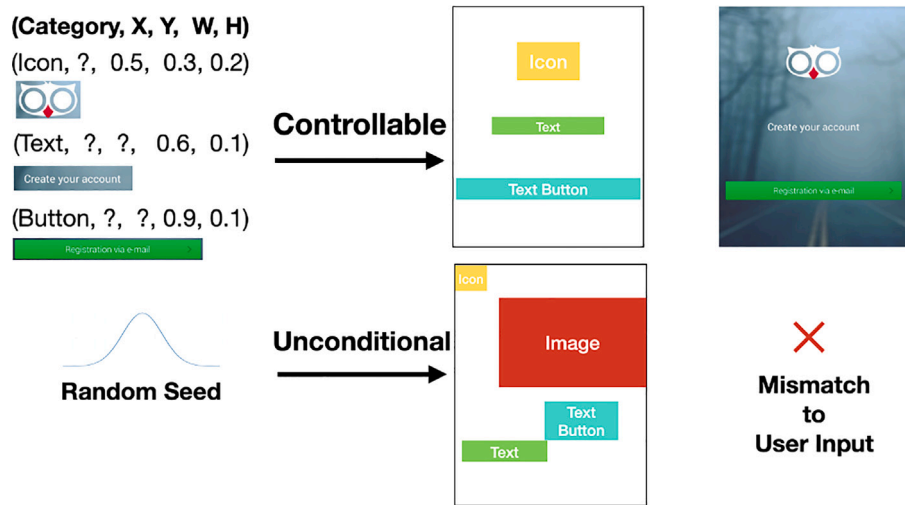


Fig. 13. Conditional (top) and unconditional (bottom) layout generation (). Each layout element is modeled by 5 attributes (Category, X, Y, W, H), representing the category, x-y coordinates, width, and height of an element, respectively. In conditional layout generation, attributes are partially provided by the user and the goal is to generate unknown attributes. While unconditional generation just produces layouts directly without considering the user requirements.

Source: Image from [23].

size and the DRL-based method outperforms the traditional method in terms of the quality of the generated results. The NDN outlined in Section 4.2.1 can also work in the advertisement design tasks, but rather than optimize the generated layouts by manual loss, it generates layouts conditioned on user-specified constraints.

4.2.3. Magazine design

Magazine design, including magazine cover design and magazine page design, typically is a combination of text and images that are arranged in a visually appealing and informative way. Most magazines are handcrafted, but it still remains challenging for small businesses and amateur individuals to design aesthetically compelling magazine layouts.

Yang et al. [79] propose a system for generating visually appealing visual-textual presentation layouts by utilizing topic-dependent layout templates created by domain experts. Zheng et al. [80] found that previous approaches mainly rely on heuristic criteria that are oblivious to the influence of content upon graphic design layouts. However, layouts should vary with visual and textual contents even within the same task category. For example, when generating a page of a fashion magazine, the layouts for the contents of interviews, brief biographies, and fashion trends are significantly different, as shown in Fig. 14. Thus, Zheng et al. make the first effort to study on content-aware layout generation model for graphic design and propose a deep generative model (ContentGAN) to capture the effect of contents on layouts and synthesize graphic design layouts based on visual and textual features without heuristic rules. The model consists of a multi-modal embedding network and a layout generative network to learn the multi-modal features and extract content-aware features. They also contribute a large-scale magazine layout dataset that benefits research on the field of the study. The visual results and user studies of the experiments are displayed but no quantitative evaluation is provided.

4.2.4. Poster design

Poster design is the practice of creating visual content that is intended to be displayed on walls or online to promote a product, service, event, or idea. Similar to advertisement design and magazine design mentioned earlier, automatic poster design has gained widespread attention from both the scientific community and the public. In addition to traditional methods such as energy-based [81,83] and template-based models [85], layout generation methods for poster design based on deep learning have become a focus of many studies.

Qiang et al. [85,86] first studied the challenging and specific issue of generating posters from original academic papers with a data-driven framework. The framework overall includes four stages, that is, (1) content extraction: extracting text content and graphical content from the paper and summarizing the textual content with the textual summary algorithms; (2) panel attributes inference: inferring key attributes for the initial panel (such as panel size and aspect ratio) by learning a probabilistic graphical model from the training data; (3) panel layout generation: synthesizing layout with the proposed recursive algorithm to further update key attributes, i.e., aspect ratio, and generating an informative and aesthetic panel layout; and (4) composition within each panel with the graphical algorithm. To validate the effectiveness of the model, the authors collect and release a new benchmark dataset, called NJU-Fudan Paper-Poster dataset, and the experiments show that the model can generate readable, informative, and visually aesthetic scientific posters from academic papers.

Li et al. [25] concentrate on another specific aspect of this task which is to generate a harmonious textual layout over an image, i.e., to position the text properly according to the requirements of the human visual system. They propose an efficient deep aesthetics learning approach with three main components, Saliency Network, Text Region Proposal, and Scoring Network. Concretely, the Saliency Network is a network based on BASNet [87] to predict the semantic visual saliency map; the Text Region Proposal is an approach with the proposed Diffusion Equation Algorithm and Text Anchor Generation Algorithm to generate some candidate text anchors; the Scoring Network consists of a multi-scale feature extraction module and an aesthetic feature extraction module to assess the aesthetic quality of the candidate textual layouts and output the optimal one. This work also contributes a new Textual Layout Aesthetics (TLA) dataset and some reasonable evaluation metrics.

Zhou et al. [21] also study poster design problems and further explore content-aware layout generation combined with spatial visual contents of input images and introduce the composition-aware graphic layout GAN (CGL-GAN) based on ContentGAN. They found that the neglect of image compositions, especially the spatial information, will ill-effect the layout results, such as occlusions on the key part of components resulting in a negative impact on subject presentations, text readability, or even the visual balance of the whole poster. To solve the problem, a Domain Alignment Module (DAM) is designed to narrow the domain gap between training data (posters) and test data (images) and obtain spatial information to help the following

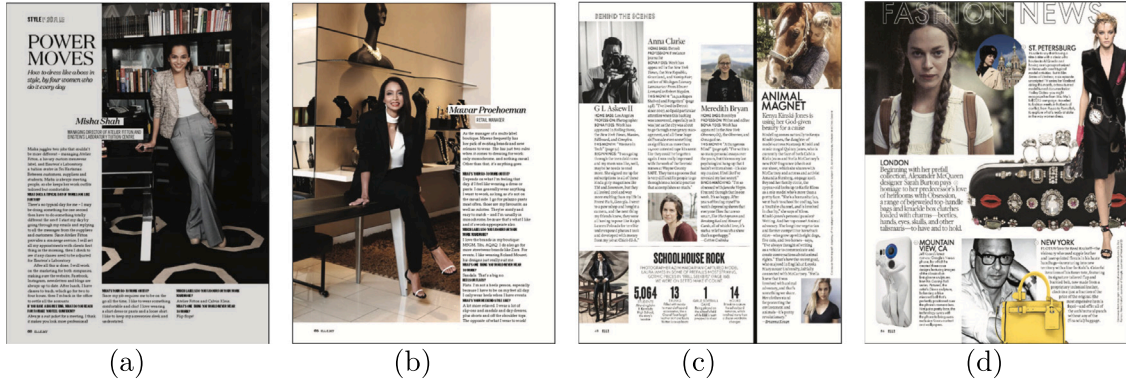


Fig. 14. Influence of content on the layout. All example pages shown here are from the fashion category: (a) and (b) on interviews; (c) on brief biographies; (d) on fashion trends. The pages are from ELLE (© Hearst Magazine Media). Source: Image from [80].

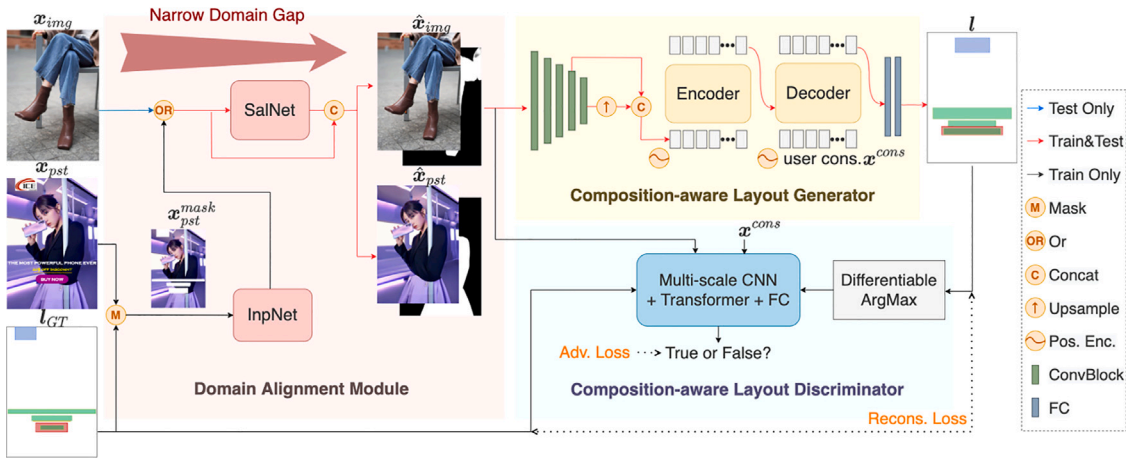


Fig. 15. The framework of CGL-GAN (image from [35]). Here, SalNet [88] is a pre-trained saliency detection net to obtain spatial information and assist the following module to better understand image compositions. InpNet [89] is a pre-trained inpainting Net to get obtain the repaired results. The CGL-GAN is trained with a reconstruction loss and an adversarial loss.

networks to better understand image compositions, and a Composition-aware Layout Generator and Discriminator are proposed to capture the relationship between image compositions and graphic layouts, as illustrated in Fig. 15.

4.2.5. Discussions

Layout generation methods for graphic design include a variety of application contexts and more than half of the literature reviewed in this paper is mainly concentrated in this domain. Different graphic design tasks share some similar design rules. For Instance, the primitives consist of image and text elements, the generated layouts are supposed to be visually balanced and attractive, and the information presentation guided by the layouts should be clear and tightly focused. Yet for different specific tasks, there are different details and focuses based on the primitives and requirements of the users. According to the introduction, the methods in this section can generate primary layouts with visual and textual primitives by employing deep generative models. To optimize the models and generate results resembling the ground truth layouts, more work is needed, which mainly can be divided into three categories: (1) adding manual designed loss [78], (2) directly providing and processing conditional auxiliary information [20,24,26,29], and (3) extracting implicit relationship or detailed information to assist the model [21]. Due to the complex relationships between primitives in graphic design tasks, the methods presented have difficulty in transferring to other specific tasks.

4.3. Document layout generation

Document layout generation can be regarded as a specialized case of graphic design, as it also involves handling both graphic and textual elements during generation, but in accordance with stricter design rules that align with the characteristics of the document. Automatic document layout generation underlies many widely used document processing applications aimed at relieving the burden on the users, such as $\text{\LaTeX}$ and Microsoft Publisher [90]. In the last decades, there has been a resurgence of interest in automatic document layout generation because of the development of information technology and multimedia. Hurst et al. [90] (2009) review the literature on early works on automatic document formatting. However, they only focused on primarily textual documents that do not involve widget layout, diagram layout, or layout issues in multimedia documents, which are the main practical application of document layout in recent years.

This section will discuss recent generic approaches related to automatic document layout generation, including pagination and page-level adjustment.

4.3.1. Pagination

Pagination is a specific document layout task in which the goal is to arrange the content properly in an optimal page count with given the content to present in a document. Damera-Venkata et al. [91] reduce this issue to probabilistic inference over the Bayesian network and propose a new paradigm for adaptive document layout generation,

namely, probabilistic document model (PDM). They expand the static templates, which encode the absolute positions of all elements by simply pasting images rather than by cropping or scaling, to probabilistic templates. The core of PDM is to encode soft constraints (aesthetic design priors) with probability distributions of template parameters on properties like whitespace, exact image dimensions, and image rescaling preferences. The model can automatically adjust the editing process to make the templates optimally fit content by computing the optimal page count and optimizing sequences of templates. PDM opens the door to leveraging probabilistic machinery and machine learning for parameterizations directly from examples in the document generation domain which may mitigate if not eliminate the burden of designing templates and specifying prior distributions.

4.3.2. Page-level adjustment

Page-level adjustment aims to generate a document presenting all content that is necessary within a page, which means no more computing for the optimal document counts compared with the pagination task. Patil et al. [37] present a novel framework to generate plausible document layouts in large quantities and varieties, which is called READ for REcursive Autoencoders for Document layout generation. Moreover, the authors introduce a combinatorial metric for measuring document similarity, DocSim (introduced in Table 4), which is used to measure the novelty and diversity of generated document layouts. Arroyo et al. [17] also adopt DocSim as a similarity metric to evaluate the generated document layouts by their proposed model VTN. READ has several limitations: (1) it is not able to reliably represent and generate highly complex layouts; (2) it has not yet considered misalignment and overlapping issues so the generated layouts may contain some undesirable artifacts. To deal with the misalignment and overlapping problem, Shi et al. [20], Cheng et al. [24], and Kikuchi et al. [26] use constraint information which can also satisfy user-specified relationship. Specifically, Shi et al. utilize manual constraints of layout primitives and facilitate L-CGAN with a graph-constraints module to optimize the generated layouts. Cheng et al. [24] attempt to adopt the recently popular diffusion models [92] to solve layout generation problems and build PPlay, a latent diffusion model with conditional guidelines. Kikuchi et al. propose a framework called CLG-LO (Constrained Layout Generation via Latent Optimization) which rely on an iterative algorithm based on the augmented Lagrangian method [93] to find a layout that satisfies the specified constraints.

4.3.3. Discussions

Document layout generation also works with visual and textual elements. However, it follows more strict design rules than the previously discussed tasks because documents focus more on alignment and overlapping. These standards are not crucial for scene layout generation and graphic design to determine the quality of the generated results. Most methods for document layout generation employ or design evaluation metrics on alignment and overlapping to quantitatively validate the effectiveness of the models. The frequently used quantitative evaluation metrics of document layouts are summarized in Section 5

4.4. Room layout generation

Room layout generation involves floor plan generation [27,28,31] and indoor scene layout generation [33,38], as shown in Fig. 16. Designing a room layout of either kind can be an expensive and time-consuming iterative process. Therefore, automated room layout generation techniques are in urgent demand with tremendous potential in the construction and real-estate industries. However, it is a great challenge to automatically generate a room layout that fulfills all the functional requirements.

In this section, we will provide a brief overview of various room layout generation methods, including floor plan generation and indoor scene layout generation.

4.4.1. Floor plan generation

Hu et al. [27] take the first step in the direction of user-guide floor plan generation and present Graph2Plan, a deep neural generative framework for automated floor plan generation that incorporates user-in-the-loop designs. The model can convert a layout graph of constraints provided by users, such as room numbers, locations, and adjacencies, along with a building boundary, into a floor plan that satisfies both the user-specified and boundary constraints. The network is trained on a large-scale floor plan dataset called RPLAN [94]. The architecture of the Graph2Plan network consists of a graph neural network (GNN), a convolutional neural network (CNN), fully-connected layers (Box), and a BoxRefineNet. Specifically, a user-constrained layout graph and building boundary are processed via the GNN and CNN, respectively, to extract features. The output features are concatenated and fed into the network Box to generate a corresponding bounding box. A cascaded refinement network (CRN) [63] follows, which can generate a floor plan image with all the predicted room boxes as inputs. Finally, the results are refined by a BoxRefineNet inspired by works on object detection [95]. The quality and versatility of Graph2Plan are verified by its performance in both qualitative and quantitative evaluations, and yet the authors point out that the model has limitations because of limited types of constraints and the scale of the dataset.

House-GAN [31] is another effective model for floor plan generation using graph constraints. Different from Graph2Plan utilizes GNN to process the layout graph, House-GAN employs convolutional message passing neural networks (Conv-MPN) [96] which allows more effective higher-order reasoning for composing layouts and validating adjacency constraints. Based on House-GAN, House-GAN++ [28] is proposed for a more challenging production-level task *i.e.*, handling non-rectangular room shapes, generating doors/entrances, and using a functional graph instead of an adjacency graph. The study also discovers that a simple non-iterative training procedure is effective in training such an iterative generator.

Graph2Plan and House-GAN require a heuristic post-process to improve the realism of the generated floor plans, which is not learned from the data. Para et al. [30] research on topologically and spatially consistent layout generation and proposed a transformer-based generative model for both floor plans and furniture layout generation. Through extensive experiments with many variations of graph-based GANs and VAEs, Para et al. found that these methods struggle with the discrete nature of graphs and layouts. Thus, they choose an autoregressive model with attention and self-attention modules that inherently addresses discrete data. The model first generates a layout graph with layout elements as nodes and constraints between layout elements as edges and solves for the spatial shape attributes for the final layout via optimization, rather than outputting shapes directly.

4.4.2. Indoor scene layout generation

Ritchie et al. [38] present an indoor scene synthesis method based on deep convolutional generative models. The model iteratively predicts the layout parameters (category, location, orientation, and size) of the indoor layout elements with a top-down-image-based representation. The elements are inserted iteratively into the scene guided by predicted parameters from four neural network modules, as shown in Fig. 17. The idea of the pipeline naturally supports both the automatic completion of partial scenes and complete indoor scene generation. Furthermore, it is experimentally confirmed that the method can complete or generate indoor scenes significantly faster than the previous methods and the result outperforms other state-of-the-art deep indoor scene generation models in terms of plausibility and perceived visual quality.

Wang et al. [33] provide insight that object-oriented approaches that infer the set of objects in a scene and their configurations excel at high-level planning tasks like floor plan generation, while space-oriented approaches perform well at instantiating a layout by placing objects in precise spatial configurations. To combine the advantages of

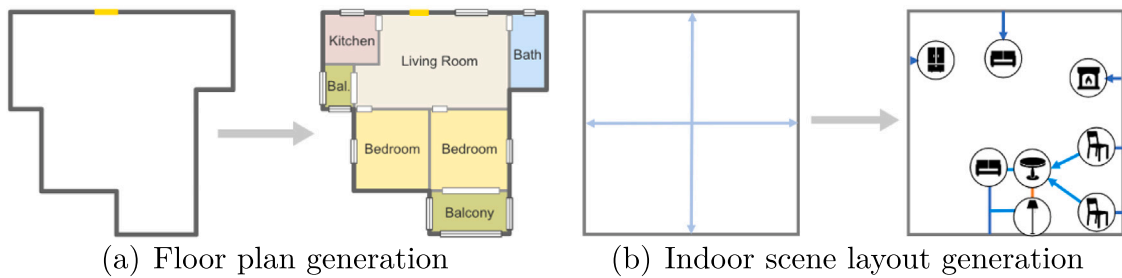


Fig. 16. Illustrations of two representative room layout generation tasks.
Source: Original images from [27,33]

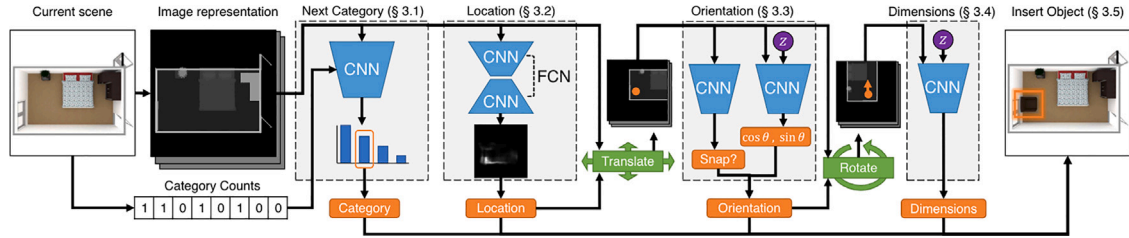


Fig. 17. Overview of the automatic object-insertion pipeline proposed by Ritchie et al.. A top-down-image-based representation of the scene is extracted and fed to a series of CNN-based decision modules to determine which category of object to add (if any), the location, orientation, and dimensions of the layout element.
Source: Image from [38]

both scene synthesis approaches, they introduce PlanIT, which utilizes a high-level relation graph representation with spatial prior neural networks. The framework decomposes the indoor scene generation task into two phases, planning and instantiation phases. In the planning phase, PlanIT encodes a scene via a relation graph and synthesizes relation graphs via a deep graph convolutional generative model. During the instantiation phase, PlanIT places indoor objects into the scene consistent with the graph by image-based convolutional network modules. With this design, the quality of the generated scenes is comparable to those generated by prior approaches in both user study and tests by learned classifiers. Additionally, it also improves the flexibility of the intermediate relationship graph representation, allowing the framework to generate scenes from a partial graph provided by users.

As the online virtual interior design is becoming increasingly demanding, it seems to grow in requirement in a shorter time. To assist designers to produce indoor design solutions at the industry level in a few seconds, Di [32] propose an assistive model consisting of the extraction of the abstract graph, conditional graph generation, and conditional scene instantiation. In particular, the extraction builds a structural graph of the indoor scenes based on geometric rules and a hierarchy rule. Then, the corpus of the extracted graphs and the users' preferences are encoded and the encoded representation is used to learn a generative model of graphs so that the generated graph is able to correspond to the representation of an indoor scene with a particular condition. Finally, the abstract conditional graph and the representation of a suggestion are encoded to instantiate into a concrete scene which is according to the choice of the professional designers and users' preferences.

4.4.3. Discussions

Room layout generation tasks closely attach to the functionality of the generated layout and users' specific demands. Both concerns are suitable to be represented via graphs and difficult to learn directly from datasets. Thus, for room layout generation, both floor plan generation and indoor scene layout generation, a graph is generally adopted to represent the layout or conditional information, and models dealing with graphs, such as GNN and GCN, are preferred. The evaluation of the performance of the models mainly relies on the demonstration of the generated layouts and statistical results of the user study. Some

study [28,31,33,38] also provide their quantitative evaluation on the layout generation model, which mainly focuses on the diversity of the layouts generated by the model, the compatibility, and inferencing efficiency.

4.5. Others

4.5.1. New-view layout synthesis

The novel view synthesis is to generate a target image given a source image, a source pose, and a target pose (see Fig. 18) which has a wide range of applications in robotics, 3D reconstruction, Augmented Reality (AR) / Virtual Reality (VR) and other fields. Existing works [98,99] mainly adopt the pixel-by-pixel prediction to synthesize target images in the image space, which tends to generate poor results with ghosting and blurry artifacts. To address this problem, Qiao et al. [22] first attempted to explore novel-view prediction in the layout space and proposed a novel-view scene layout generation model. The main difference between their work and previous works is illustrated in Fig. 19. Previous works directly map the input layout to the 3D scene representation (e.g., MPI [99] or hybrid representation [98]), and then perform pixel-wise projection, while Qiao et al. proposed an approach that considers spatial and semantic interactions among objects in the layout space without explicit 3D modeling.

New-view layout synthesis not only enables generating consistent layouts from different views even with large camera movements, but also provides scene understanding priors for a wide range of applications, including novel-view image synthesis, novel-view image editing, and amodal object estimation.

4.5.2. 3D shapes

Language-based 3D scene generation is introduced as a new paradigm to let ordinary users create 3D scenes effortlessly and immediately [100]. The main technical route to generate 3D shapes is to convert the semantic representation to a set of low-level depicitors representing 3D objects, poses, spatial relations, color attributes, etc. Although 3D shapes generation has been studied for decades, generating 3D shapes from a layout perspective has not been proposed until recently researchers proposed that spatial knowledge is useful for text-to-3D scene generation [101]. Resolving conflicts and inferring

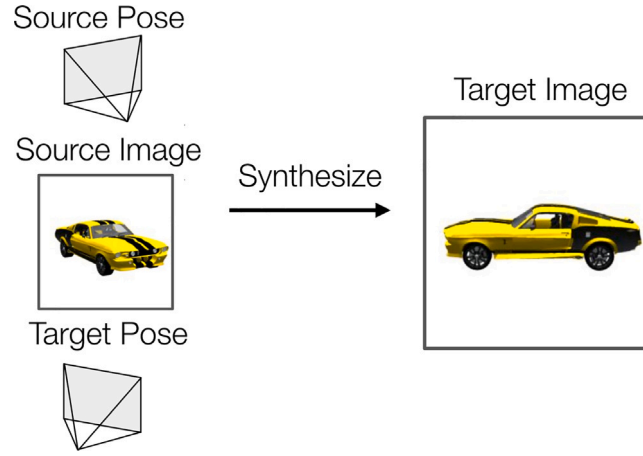


Fig. 18. Novel view synthesis.
Source: Image from [97]

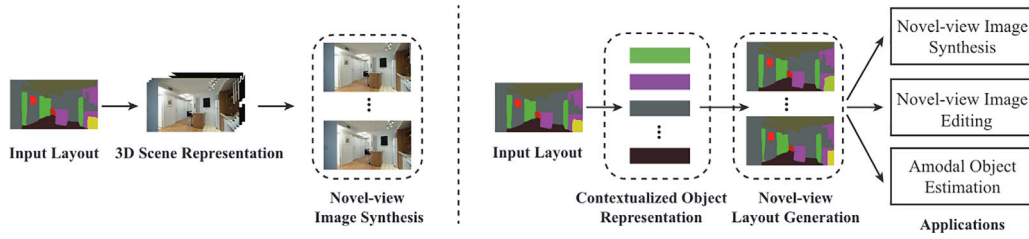


Fig. 19. A brief comparison between existing novel-view layout-to-image synthesis methods [98,99](left) and the proposed novel-view scene layout generation method (right). Note that different from the previously introduced layout generation tasks, the layouts of new-view layout synthesis are semantic maps rather than bounding boxes.
Source: Image from [22].

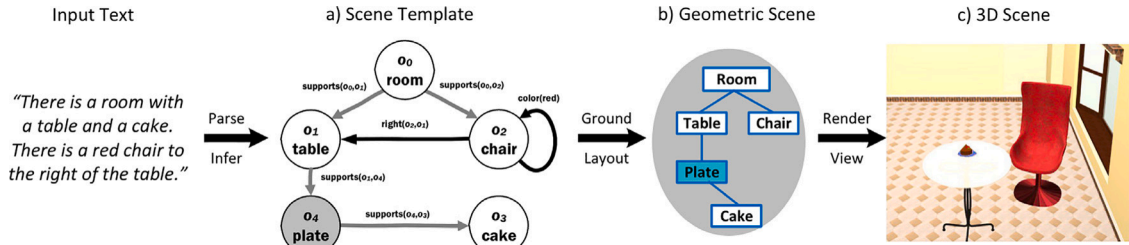


Fig. 20. Overview of our spatial knowledge representation for text-to-3D scene generation.
Source: Image from [101].

implicit constraints is critical for achieving plausible results of 3D shape generation. Chang et al. [101] present a spatial knowledge representation learned from 3D scenes and model spatial relations between language and geometric constraints. By feeding parsed text into a scene template, implicit spatial constraints are inferred from learned priors, then the template is grounded to a geometric scene to instantiate and arrange a final 3D scene. The overall pipeline is depicted in Fig. 20, and the authors believe that spatial knowledge is critically useful not only in 3D shape generation tasks but also in other domains where the understanding of pragmatics is necessary.

Inspired by the nature of human perception of 3D shapes as a collection of simple parts, which is consistent with the characterization of object perception provided by recognition-by-components (RBC) [102], Zou et al. [39] explore an abstract shape representation based on primitives and propose a model based on a generative recurrent neural network called 3D-PRNN. Although 3D-PRNN has fewer degrees of freedom in representation, it achieves competitive accuracy with voxel-based reconstruction methods. Sung et al. [40] propose

a weakly-supervised model for retrieving and suggesting complementary components with their placement for a partial 3D assembly. The model does not rely on consistent segmentation or part labeling but on unlabeled data, which naturally allows it to deal with ambiguities and seamlessly integrate user preferences into the design process. By jointly training embedding and retrieval networks, the network learns how to predict complementary parts and how to organize them in a low-dimensional manifold, and instead of predicting a single output, the network predicts a probability distribution over the space of part embeddings. Wu et al. [103] present a sequence-to-sequence (Seq2Seq) autoencoder PQ-NET, which learns 3D shape representations as a sequential part assembly. PQ-NET can not only be adapted to random 3D shape generation but also generative tasks such as single-view 3D reconstruction and shape completion. LayoutTransformer [16] follows a procedure similar to PQ-Net and trains an autoencoder with self-attention blocks [6] to learn contextual relationships between layout primitives and generate novel layouts in diverse domains including 3D shape generation.

Table 3

A summary of the public datasets involved in the deep layout generation models discussed in this paper.

Tasks	Datasets	Models	Backbone
MNIST digit layout	MNIST-Layouts ^a [34]	LayoutVAE [34]	VAE [7]
MNIST digit dot layout generation	MNIST Layouts ^a [15]	LayoutGAN [15] L-CGAN [20]	GAN [9] CGAN [104]
Natural scene layout generation	COCO [105]	LayoutVAE [34] LayoutTransformer [16] VTN [17] LT-Net [36] BLT [23]	VAE [7] Transformer [6] Transformer [6] Transformer [6] Transformer [6]
		Visual Genome [106]	LT-Net [36] Transformer [6]
		Abstract Scene Dataset v1.1[107]	LayoutGAN [15] L-CGAN [20]
Advertisement design	Image banner ads ^a [29]	NDN [29]	GNN [108]
	CGL-Dataset ^a [21]	CGL-GAN [21]	GAN [9]+CNN [109]+Transformer [6]
Magazine layout generation	MagLayout ^a [80]	ContentGAN [80] BLT [23]	VAE [7]+GAN [9] Transformer [6]
UI design	RICO [110]	NDN [29] VTN [17] LayoutTransformer [16] BLT [23]	GNN [108] Transformer [6] Transformer [6] Transformer [6]
		LayoutGAN [15] LayoutTransformer [16] VTN [17] L-CGAN [20] BLT [23] PLAY [24]	GAN [9] Transformer [6] Transformer [6] CGAN [104] Transformer [6] Diffusion model [92]
		ICDAR2015[112]	READ [37] RvNN [113]+VAE [7]
Room layout	RPLAN [94]	Graph2Plan [27] Para et al. [30] House-GAN++[28]	GNN [108]+CNN [109] Transformer [6] GAN [9]
	SUNCG [114]	PlanIT [33] Ritchie et al. [38]	GCN [115] CNN [109]
	SUN RGB-D [116]	VTN [17]	Transformer [6]
3D shape generation	LIFULL HOME [31]	House-GAN [31]	Conv-mpn [96]+GAN [9]
	PartNet [117]	LayoutTransformer [16]	Transformer [6]
	ModelNet [118]	3D-PRNN [39]	RvNN [113,119]
Novel view layout synthesis	NYU Depth V2 dataset [120]	3D-PRNN [39]	RvNN [113,119]
	RealEstate10K [121]	Qiao et al. [22]	Attention mechanism [6]

^aIndicates the dataset is originally proposed for layout generation tasks.

4.5.3. Discussions

Methods for layout generation problems have provided a new idea for many well-studied issues, including 2D scene generation (text-to-image generation, novel view synthesis, etc.) and 3D shape generation. The mainstream approaches for 2D scene generation and 3D shape generation are directly generating pixel-wise and voxel-wise results, respectively. In recent years, by combining layout generation methods some researchers adopt a two-stage strategy that generates task-oriented layouts in the first step and further generates the final pixel-wise or voxel-wise results. The effectiveness of the coarse-to-fine strategy has been proved by competitive performance in diverse domains [16,22,36,39,103].

4.6. Review and summary

In this survey, we introduce studies on layout generation from various categories, depending on the application area. Most of them emphasize the importance of the primitive relationships contained in a layout. In essence, all approaches aim to address the core problem (i.e., generating layout) and specified downstream tasks. Note that requirements for generated layouts differ oriented by the downstream tasks. Table 3 summarizes the public datasets involved in the key advances in this field through deep learning methods. The datasets have

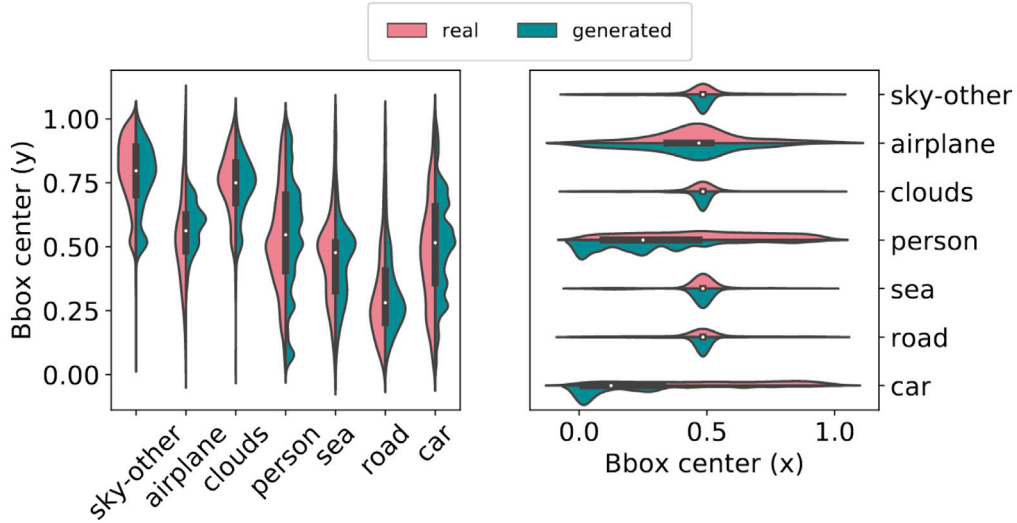
been categorized in accordance with the layout generation tasks. The models that have been trained on the dataset are also summarized in the table followed by their backbone models.

5. Evaluation methodology

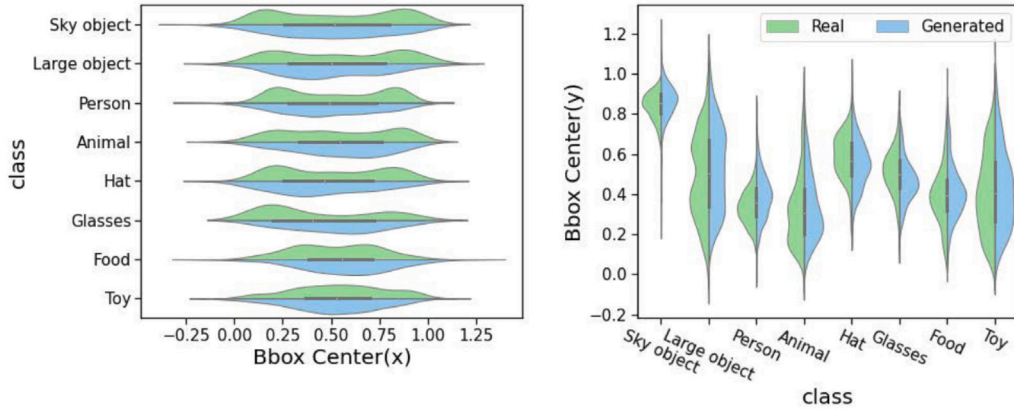
In general, there are two types of evaluation methodologies for assessing the performance of the models i.e., qualitative evaluation and quantitative evaluation. Qualitative evaluation in this field relies on the subjective interpretations of visual results and aesthetic judgments of observers. In parallel, quantitative evaluation is focused on reproducible and reliable evaluation metrics, which can be used to develop a standard of practice. In this section, we review evaluation methodologies used in different tasks to assess the models in both qualitative and quantitative terms.

5.1. Qualitative evaluation

To qualitatively evaluate the advantages of the models, it is common to present the results in a visual way, which makes it intuitive and widely adopted, as well. Due to space limitations, the results presented visually in a paper are limited, which sometimes can hardly represent the overall performance level.



(a) Distribution of xy-coordinates of the centers of the Bounding boxes generated by LayoutTransformer[16] trained on COCO and real ones(original images from [16]).



(b) Distribution of xy-coordinates of the centers of the Bounding boxes generated by L-CGAN[20] trained on AbstractScene-Layouts and real ones(original images from [20])

Fig. 21. Distribution of xy-coordinates of the centers of real Bounding boxes and those generated by LayoutTransformer and L-CGAN, respectively.

There are some other alternatives to qualitatively evaluate the performance of the layout generative models. Gupta et al. [16] and Shi et al. [20] visualize the distribution of xy-coordinates of generated Bounding boxes compared with the real distribution. The comparison depicted by Fig. 21 indicates that their proposed layout generative models have successfully captured the distribution of the ground truth, thus reflecting the model's performance to an extent. Gupta et al. also prove LayoutTransformer captures contextual relationships between various elements, by visualizing the 2D-tSNE plot of the learned embeddings for categories of COCO. As shown in Fig. 22, embeddings are colored and labeled according to their categories, and semantically similar categories cluster or close together. For instance, cats are closer to dogs than sheep or cows.

5.2. Quantitative evaluation

Quantitatively evaluating generative models has always been a delicate matter, especially for layout generation models, due to the variety of downstream tasks involved. Quantitative evaluation varies between different layout generation studies. In this section, we will discuss the quantitative evaluation methods adopted in the prior work.

5.2.1. User study

Many studies, *e.g.*, literature [15,20,24,78,80], implement user study, *i.e.*, human evaluation, for extensive subjective evaluation.

Specifically, each user will be asked to rank the randomly sampled results from different models, or from the real samples, according to the given criteria. The ranking results are generally depicted in histograms, tables, or other visualizations. The user groups can be average users or professional users with rich design experience, also the user study can be conducted via Amazon Mechanical Turk.

5.2.2. Toy experiment

To validate the effectiveness of the abstract scene layout generation models, Li et al. [15] and Shi et al. [20] implement a toy experiment on dataset MNIST layouts [15] where each handwriting digit is represented as a set of coordinates of 128 points. Concretely, by generating the layout of 128 points, MNIST digit images can be formulated. For quantitative evaluation, a digit classifier is pre-trained and used to recognize randomly sampled generated MNIST digits. The performance of the model can be evaluated by comparing the recognition accuracy and Inception Score (IS) [122] of the generated results. IS is the most widely used metric in generative modeling for images to codify the sharpness and diversity [123]. It is calculated according to the following objective:

$$\text{IS}(G) = \exp(\mathbb{E}_{x \sim p_g} D_{KL}(p(y|x) \| p(y))), \quad (11)$$

where $x \sim p_g$ denotes the image generated from the Generator, $p(y|x)$ is the conditional label distribution, and D_{KL} is the Kullback-Leibler

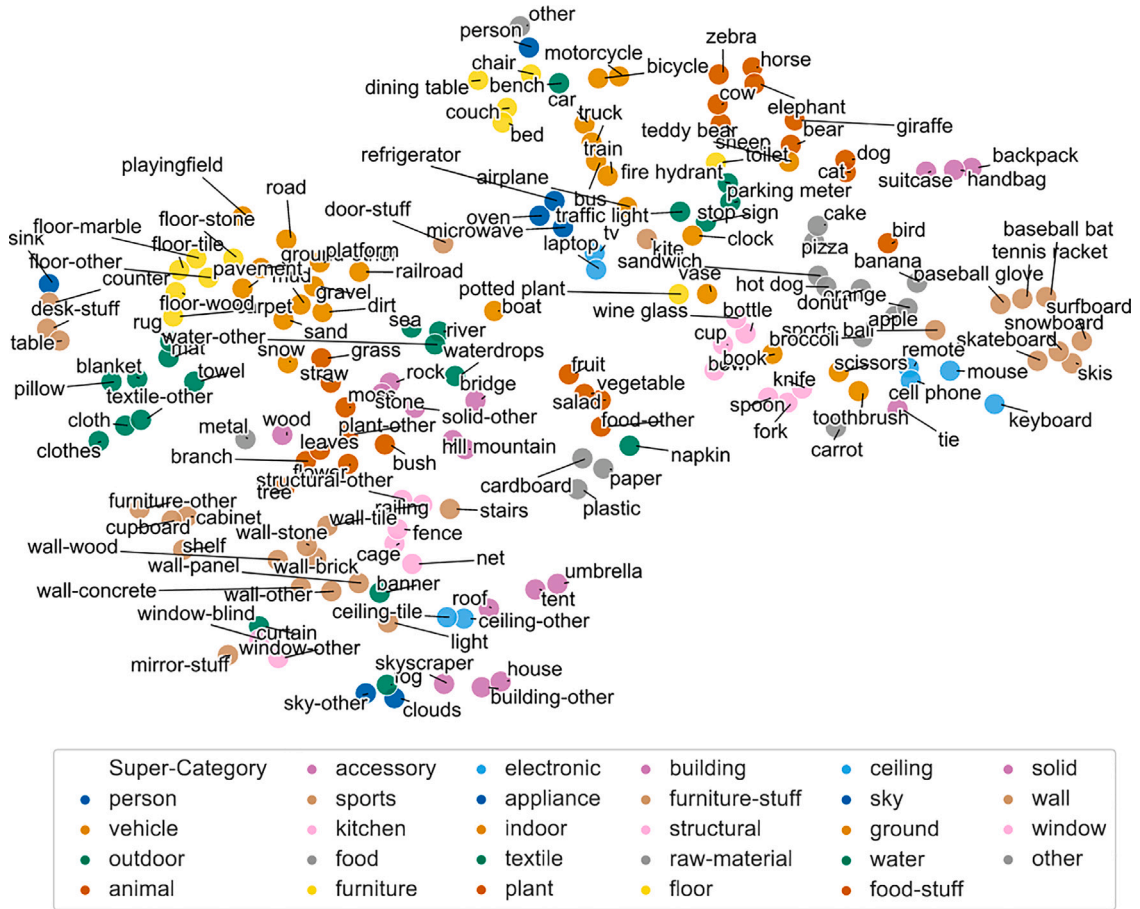


Fig. 22. TSNE plot of learned category embeddings.
Source: Original image from [16].

divergence between $p(y | x)$ and $p(y)$, computing as follows:

$$D_{KL}(P \parallel Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}. \quad (12)$$

5.2.3. Fidelity

The fidelity of generated layout results is usually measured by following metrics, especially for natural scene layout generation tasks. Noticeably, some metrics (e.g., FID, NLL, IS) cannot directly implement on layouts but on the downstream task results, that is the final scene images guided by the generated layouts. Thus, it usually needs to employ layout-to-image models such as Layout2Im [75] to obtain the synthesized scene images.

Fréchet inception distance (FID). FID is initially defined to compute the distance between two multivariate normal distributions [124]. In the context of computer vision, Heusel et al. [125] apply FID to evaluate the performance of generative models, especially GAN, and experimentally validate that it is more consistent than the Inception Score. For layout generation, FID also works so that it is adopted in many relative works [16,29,36]. Specifically, FID captures the similarity of generated images to real ones following the given objective:

$$FID = \|\mu_X - \mu_Y\|^2 - \text{tr}(\sum_X + \sum_Y - 2\sqrt{\sum_X \sum_Y}), \quad (13)$$

where X and Y are the real and fake embeddings from the Inception model [126], respectively. μ_X and μ_Y are the mean of X and Y . $\sum_X$ and $\sum_Y$ are the covariance matrix of the embeddings and tr is the trace of the matrix.

Negative log-likelihood (NLL). In the computer vision community, NLL was initially implemented for multiclass classification problems,

calculated through the given Eq. (14). Note that the vector $\hat{y}$ represents a discrete probability distribution over the possible values of the observation and the vector y can also be interpreted as a probability distribution over the same space. Ideally, the average NLL of all the layouts in validation data can be utilized to evaluate the performance of a generative model on the layout generation downstream task.

$$NLL = -\sum_{j=1}^M y_j \log \hat{y}_j. \quad (14)$$

Mean intersection over union (MIoU). Intersection over Union (IoU) is the most popular evaluation metric for tasks such as segmentation, object detection, and tracking by computing the intersection over the union of the ground truth bounding box and the prediction bounding box. It can be applied for layout generation tasks to measure how well the generated layout matches the ground truth bounding box. It is calculated as follows:

$$MIoU = \frac{1}{k} \sum_k \frac{G \cap R}{G \cup R}. \quad (15)$$

Here G denotes the generated layout bounding box and R represents the real (ground truth) bounding box.

KullbackLeibler Divergence (D_{KL}). Ritchie et al. [38] utilize KullbackLeibler divergence (D_{KL}) to assess if an indoor scene generation model can capture the training data well. As defined in Eq. (12), D_{KL} for this context is formulated as follows:

$$D_{KL}(P_{synth} \parallel P_{dataset}) = \sum_i P_{synth} \log \frac{P_{synth}}{P_{dataset}}, \quad (16)$$

where P_{synth} and $P_{dataset}$ represent the category distribution of synthesized results and that of the training set, respectively.

Generation accuracy. For experimental measurements of indoor scene layout generation, the accuracy of the graph generation for each type of rooms is defined by [32]:

$$ACC_G = \frac{1}{N_c} \frac{\sum_{k=q}^n \mathbf{label}(g_k) = i}{N_i}, \quad (17)$$

where N_i is the number of generated graph for label i and N_c is the number of the category of the graph.

Classification accuracy. In order to demonstrate the capability of the indoor scene generation model, Ritchie et al. [38] and Wang et al. [33] opt to train a classifier to distinguish real scenes (from the training set) from generated scenes and calculate the classification accuracy. As opposed to classification tasks where higher classification accuracy yields better performance of the model, lower classification accuracy (closer to 50%) in this task implies that it is harder to distinguish between real scenes and generated ones, which in turn reflects the power of the generative model.

Chamfer Distance (CD). Generally, 3D shapes are represented by point clouds. Chamfer Distance is a nearest-neighbor-based method and is defined to compare two point clouds S_1 and S_2 as follows:

$$d_{CD}(S_1, S_2) = \frac{1}{|S_1|} \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2 + \frac{1}{|S_2|} \sum_{y \in S_2} \min_{x \in S_1} \|y - x\|_2. \quad (18)$$

Each point $x \in S_1$ finds its nearest neighbor in S_2 and vice versa; all the point-level pair-wise distances are averaged to produce the shape-level distance.

Earth Mover's Distance (EMD). Earth Mover's Distance is another metric for comparing two 3D shapes with point clouds and it is defined as:

$$d_{EMD}(S_1, S_2) = \min_{\phi: S_1 \rightarrow S_2} \sum_{x \in S_1} \|x - \phi(x)\|_2. \quad (19)$$

The metric relies on finding a one-to-one bijection mapping $\phi: S_1 \rightarrow S_2$ and it applies when $|S_1| = |S_2|$. The pairwise distances are then calculated between x and $\phi(x)$.

5.2.4. Visual balance

Visual balance usually means a balanced distribution of elements, which is essential for graphic design. Typically, there are two types of balance: vertical (top-bottom) balance and horizontal (left-right) balance. Their values can be formulated as [18]:

$$e_{balance}^{l-r} = \frac{\min(A_l, A_r)}{\max(A_l, A_r)}, \quad (20)$$

$$e_{balance}^{t-b} = \frac{\min(A_t, A_b)}{\max(A_t, A_b)}, \quad (21)$$

$$E_{balance}^{l-r} = E(e_{balance}^{l-r}), \quad (22)$$

$$E_{balance}^{t-b} = E(e_{balance}^{t-b}). \quad (23)$$

Where A_l is the overlap area between design elements and the left 1/3 part of a layout, similarly, A_r , A_t , and A_b are the overlap areas between design elements and the right, top, and bottom 1/3 part of a layout, respectively. A layout is divided equally into three parts in horizontal and vertical directions when computing these terms.

The visual balance evaluation score can be defined as:

$$E = a_1 * E_{alignment} + a_2 * E_{overlap} + a_3 * E_{balance}^{l-r} + a_4 * E_{balance}^{t-b}, \quad (24)$$

where $E_{alignment}$ and $E_{overlap}$ are computed as follows:

$$e_{align}^a = \frac{\sum_{i \in (all)} \sum_{j \in (all)} I_{ij}^a}{n^2}, \quad (25)$$

$$E_{alignment} = \max(E(e_{align}^a)). \quad (26)$$

$$e_{overlap} = 1 - \frac{\sum_{i=1}^n A_{overlap_i}}{wh}, \quad (27)$$

$$E_{overlap} = E(e_{overlap}). \quad (28)$$

5.2.5. Readability

Order Loss. The reading order of the elements is commonly expected to follow the left-to-right and top-to-down observation order, which means the upper-left corner of the layout canvas is expected to have a prior reading order. To generate a layout with plausible reading order, a predefined reading order of each element is necessary and an Order loss is introduced in literature [78], as shown in the following objective:

$$L_{ord} = \sum_{i=1}^N \sum_{j=1}^N \mathbf{1}[o_i < o_j] k(d_i - d_j), \quad (29)$$

where o_i is denoted as the predefined reading order of element i , $k(x) = \max(0, x)$, $\mathbf{1}[o_i < o_j]$ equals 1 when $o_i < o_j$ and 0 otherwise, and d_i its distance to the layout origin which is calculated as $d_i = \sqrt{x_i^L^2 + y_i^L^2}$, where (x_i^L, y_i^L) represents the upper-left coordinates of the predicted layout bounding box.

Margin Area Loss. To control the area of the generated bounding box for different graphic design elements, Li et al. [78] feed the expected area attributes to the generator and introduce a margin area loss:

$$L_{area} = \sum_{i=1}^N k\left(\frac{|s'_i - s_i|}{s_i} - \alpha\right), \quad (30)$$

where s'_i and s_i are denoted as the area of the predicted bounding box and the input expected area for element i respectively, and they are expected to be close in value while remaining some degree of flexibility to better balance different design principles for layout optimization. $k(x) = \max(0, x)$ is implemented as ReLU activation, and α is a hyper-parameter to control the allowable relative area difference.

Overlapping Rate. The overlap is supposed to be avoided in document layouts and graphic design tasks since it can have a negative impact on information presentation. Intuitively, the value of the overlap between elements is calculated as:

$$E_{Overlap} = \sum_{i=1}^N \sum_{j \neq i} \frac{s_i \cap s_j}{s_i}, \quad (31)$$

where $s_i \cap s_j$ denotes the overlapping area between the bounding boxes of elements i and j .

Alignment Rate. The alignment of layouts is usually essential to provide visual connectivity, especially in document layout problems. It requires the edges of primitives to be lined up usually in six possible alignment types: Left, X-center, Right, Top, Y-center, and Bottom aligned, which can be denoted as $\theta = (x^L, x^C, x^R, y^T, y^C, y^B)$. Take the example in literature [18], the value of the alignment can be defined as:

$$e_{align}^\theta = \frac{\sum_i \sum_j I_{ij}^\theta}{n^2} \quad (32)$$

$$E_{Alignment} = \max(E(e_{align}^\theta)), \quad (33)$$

where a sigmoid function is used to reshape an evaluation term $e(X)$ as $E(e(X)) = \arctan(e(X)\alpha) / \arctan(\alpha)$, where the larger value of α makes $E(e(X))$ more sensitive to small changes and I_{ij}^θ is the fraction of element i and j in θ -aligned manner.

Lee et al. [29] formulate the alignment metric as follows:

$$alignment_{gen} = \frac{1}{N} \sum_k \sum_i \min_{j, i \neq j} \left\{ \min \left(l(e_i^k, e_j^k), m(e_i^k, e_j^k), r(e_i^k, e_j^k) \right) \right\}. \quad (34)$$

5.2.6. Diversity

Learned Perceptual Image Patch Similarity (LPIPS). The LPIPS [127] is a perceptual metric for assessing the perceptual similarity between two images. Since diversity and similarity are negatively correlated, LPIPS can also be used to measure the diversity

of the generated images and Yang et al. [36] use LPIPS as Diversity Score(D.S.). Specifically, it computes the distances between the ground truth x and generated image x_0 pairs at feature space following the equation below:

$$d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (\hat{y}_{hw}^l - \hat{y}_{0hw}^l)\|_2^2, \quad (35)$$

where $\hat{y}_{hw}^l, \hat{y}_{0hw}^l \in \mathbb{R}^{H_l \times W_l \times C_l}$ indicate the feature stack extracted from L layers and unit-normalize in the channel dimension, respectively. The activations are scaled channel-wise by vector $w_l \in \mathbb{R}_l^C$, then the ℓ_2 distance is computed. Lastly, the spatial average channel-wise sum is calculated. Note that using $w_l = 1/l$ is equivalent to computing cosine distance.

DocSim. Consider a bipartite graph where one part contains all boxes of D_1 while the other part consists of all boxes of D_2 , and the edge weight $W(B_1, B_2)$ for $B_1 \in D_1$ and $B_2 \in D_2$ is as described above. A maximum weight matching $M(D_1, D_2)$ in this bipartite graph using the well-known Hungarian method [128]. The similarity score between D_1 and D_2 is defined as

$$DocSim(D_1, D_2) = \frac{1}{M(D_1, D_2)} \sum W(B_1, B_2), \quad (36)$$

where the sum is over all pairs $(B_1, B_2) \in M(D_1, D_2)$.

5.2.7. Compatibility

View Semantic Consistency (VSC). It is important for the novel view layout synthesis task to generate layouts of the target view from the same source to be consistent in terms of the object semantics. Therefore, Qiao et al. [22] define a new metric, namely View Semantic Consistency (VSC), to measure such consistency as:

$$VSC = \|W_s(X_1^1) - W_s(X_2^2)\|, \quad (37)$$

where X_1^1 and X_2^2 represent the generated scene layouts from two user-specified target poses. W_s is utilized to warp the generated layout back to the source pose through the depth information derived from MiDaS [129].

Graph edit distance (GED). In room layout generation [28,31], the compatibility is measured by graph editing distance [130] between the input bubble diagram and the bubble diagram constructed from the output layout in the same way as the GT preparation above.

5.2.8. Comparative metric

Quantitative evaluation methods for layout generation, which were introduced earlier, are absolute metrics that can be interpreted differently depending on the type of datasets or application contexts. For example, a lower overlapping rate could be preferred in the document layout task, while in scene layout generation the overlapping rate has no general objective value. A similar situation exists with MIOU, Alignment Rate, Order loss, and other absolute metrics. Therefore, a comparative metric is proposed to assess how close a given absolute metric of the generated results $value_{gen}$ is to that of the grounding truth $value_{real}$ [131]. Simply computing the ratio between two absolute metric values would be a straightforward comparison, but it is a problem if the divider is equal to zero. To avoid such an issue, the comparative metric has been formulated as follows:

$$comp(value_{gen}, value_{real}) = \left| \log \frac{value_{gen} + 1e - 10}{value_{real} + 1e - 10} \right|. \quad (38)$$

5.2.9. Efficiency

Inference efficiency. Comparing the average time taken to generate a required layout across different methods with the same device can reflect the performance of the models. The shorter the average time it cost, the better the model is. For room layout generation [38], speed comparisons are adopted to evaluate the layout generative models quantitatively.

5.3. Review and summary

A summary of the corresponding quantitative evaluation methodology can be found in Table 4. Noted that “↑” means the larger the better; “↓” means the smaller the better.

6. Conclusion and future directions

Layout generation is intriguing the computer vision community because of its remarkable performance and tremendous potential. Primarily, it offers a novel concept for many visual tasks, that is, disentangling a task at the layout level first and enhancing the results dependent upon inferred layouts which define a structure based on object instances and provide fine-grained information, such as the number of objects, object category, location, size, shape, etc. Recall how the human cognitive system works in generative tasks, and it can be noted that human beings have an aptitude for working in a coarse-to-fine manner, i.e., using prior knowledge to generate layout-wise before adding details later. For example, when sketching, the outline of the image is formulated based on geometric rules and conceptual links to ensure accuracy and proportion, followed by coloration and texture. The research of layout generation problems drives the implementation of a more advanced coarse-to-fine workflow. To make use of the potency of layout generation, as outlined in this survey, a number of approaches have been proposed in recent years. These methods have been discovered to be effective for generative tasks, including scene generation, document generation, graphic design, room layout generation, novel view layout synthesis, and 3D shapes generation. Nevertheless, the potential of layout generation has not yet been fully explored, and there are still several challenges that need to be resolved. Here, we discuss these challenges and provide insights into future prospects.

Challenges. The challenges in layout generation are included but not limited to the following points: (1) The representation of the layouts is inconsistent and it varies in different models even on the same task. So far, it has not been proven which representation format is most suitable for a certain task. (2) The representation of the relationships between the primitives, both spacial and semantic, is challenging to extract precisely. Although models based on Transformer, attention, and self-attention [6] mechanism show good performance on learning the relationships between the primitives, they fail to capture some fine but important relationships. (3) For graphic design tasks, it is necessary to evaluate the results from an aesthetic perspective. However, There is still a long way to design a proper quantitative aesthetics assessment. (4) The generalization and robustness of layout generation models remain open problems. Compared to image generation, layout generation tasks generally need to comply with more rules, such as design rules and user-specific constraints. Consequently, the nature of tasks and datasets has a significant impact on the generalization and robustness of layout generation models. It is a difficult problem to promote the generalization and robustness of a layout generation model.

Future Prospects. Although motivated by both scientific challenges and industrial demands, researchers have studied layout generation issues and proposed many models, these works still have much room for improvement. Apart from addressing the aforementioned challenges, future prospects in layout generation include two aspects: enriching the capability of layout generation models and combining layout generation methods with tasks in other domains. Concretely, the capability of layout generation models can extend to considering the design intent, respecting a particular design style, choosing adaptive fonts, etc. On the other hand, based on the promotion of applying layout generation methods to natural scene synthesis, novel view synthesis, and 3D shapes generation, it is of high research interest to combine layout generation methods with other layout-determined tasks such as style transfer and image inpainting.

Table 4
Evaluation Methodology.

Tasks	Evaluation perspectives	Evaluation metrics	Criterion	Ref
All tasks	User study	Users' ratings	Rating rules	–
Scene layout	Sharpness & Diversity	$IS(G) = \exp(\mathbb{E}_{x \sim p_g} D_{KL}(p(y x) p(y)))$	↑	[15,20,35]
	Fidelity	$FID = \ \mu_X - \mu_Y\ ^2 - tr(\sum_X + \sum_Y - 2\sqrt{\sum_X \sum_Y})$	↓	[16]
		$MIOU = \frac{1}{k} \sum_{k=1}^k \frac{G \cap R}{G \cup R}$	↓	[34]
		$NLL = -\sum_{j=1}^M y_j \log \hat{y}_j$	↓	[34]
Graphic design	Diversity	$d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \ w_l \odot (\hat{y}_{hw}^l - \hat{y}_{0hw}^l)\ _2^2$	↑	[36]
	Margin Area Loss	$L_{area} = \sum_{i=1}^N \mathbf{k}(\frac{ s'_i - s_i }{s_i} - \alpha)$	↓	[78]
	Order Loss	$L_{ord} = \sum_{i=1}^N \sum_{j=1}^N \mathbf{1}[o_i < o_j] \mathbf{k}(d_i - d_j)$	↓	[78]
	Visual balance	$E = a_1 * E_{alignment} + a_2 * E_{overlap} + a_3 * E_{balance}^{t-r} + a_4 * E_{balance}^{t-b}$	↓	[18]
Document layout	Readability	$alignment_{gen} = \frac{1}{N} \sum_i \sum_{j,i \neq j} \min(l(e_i^k, e_j^k), m(e_i^k, e_j^k), r(e_i^k, e_j^k))$	↑	[29]
		$E_{Overlap} = \sum_{i=1}^N \sum_{j \neq i} \frac{s_i \cap s_j}{s_i}$	↓	[78]
	Diversity	$DocSim(D_1, D_2) = \frac{1}{M(D_1, D_2)} \sum W(B_1, B_2)$	↑	[37]
Novel view layout synthesis	Compatibility	$VSC = \ W_s(X_1^1) - W_s(X_1^2)\ $	↑	[22]
Room layout	Plausibility	$MIOU = \frac{1}{k} \sum_{k=1}^k \frac{G \cap R}{G \cup R}$	↓	[27]
		$D_{KL}(P_{synth} P_{dataset}) = \sum_i P_{synth} \log \frac{P_{synth}}{P_{dataset}}$	↓	[38]
		Classification accuracy	↓	[38]
		$ACC_G = \frac{1}{N_c} \sum_{i=1}^N \frac{\mathbf{1}[label(s_i)=i]}{N_i}$	↑	[32]
	Diversity	$FID = \ \mu_X - \mu_Y\ ^2 - tr(\sum_X + \sum_Y - 2\sqrt{\sum_X \sum_Y})$	↓	[16]
	Compatibility	GED [130]	↓	[28,31]
	Inference efficiency	Average time taken to generate a required layout with the given device	↓	[38]
3D shapes	Plausibility	$d_{CD}(S_1, S_2) = \frac{1}{ S_1 } \sum_{x \in S_1} \min_{y \in S_2} \ x - y\ _2 + \frac{1}{ S_2 } \sum_{y \in S_2} \min_{x \in S_1} \ y - x\ _2$	↓	[16]
		$d_{EMD}(S_1, S_2) = \min_{\phi: S_1 \rightarrow S_2} \sum_{x \in S_1} \ x - \phi(x)\ _2$	↓	[16]
–	Comparative metric	$comp(value_{gen}, value_{real}) = \log \frac{value_{gen} + 1e-10}{value_{real} + 1e-10} $	↓	[131]

CRedit authorship contribution statement

Yong Shi: Funding acquisition, Resources, Conceptualization, Supervision. **Mengyu Shang:** Writing – original draft, Visualization, Methodology. **Zhiqian Qi:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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